SECTION 3 – SERVICE

3–1 INSTALLATION

Install Lightning PDU using the procedures of this section. Complete installation in accordance with NEMA standards and local electrical code as appropriate.

3–1–1 Unpacking and Positioning

- The Lightning PDUs are shipped on integral casters, wrapped in plastic.

Note

If shock watch indicator is red, Lightning PDU may have been damaged during shipment. Notify carrier immediately.

1. Remove cardboard cover.
2. Roll to installation site on integral casters.

CAUTION

Lightning PDU weighs about 1000 lbs (455 kg). Use care when rolling Lightning PDU. It weighs 250 lbs (114 kg) per caster. Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.

3. Remove protective plastic film.
4. Before final positioning of Lightning PDU, verify that all cable routing conduit and raceway is positioned properly. If Lightning PDU final position does not provide access to back or a side, route installation cables into cabinet before final positioning.

Note

For installation, 3 ft. (91 cm) of unobstructed space is needed for access to one side or back of Lightning PDU. Access through front door is only access needed for periodic maintenance and repair.

3−1−1 Unpacking and Positioning (continued)

5. Move Lightning PDU to its final position.

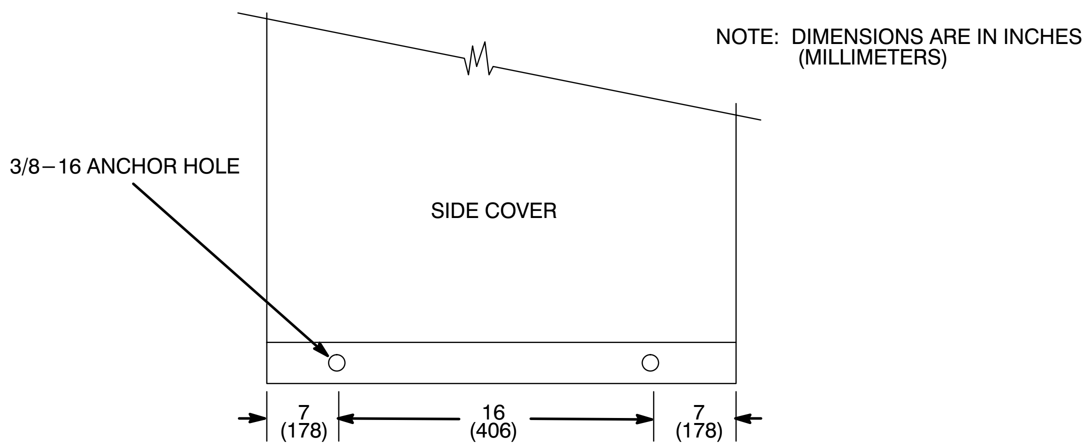
6. For a mobile installation, or if anchoring is required by local seismic code:

   a. Secure to floor using floor anchors.

      – Mark floor anchor prior to drilling. Hold floor anchor securely to base of Lightning PDU. Using transfer bolt in documentation container inside outer door, inside Lightning PDU strike transfer bolt through each Lightning PDU anchor hole to mark floor anchor. See Illustration 3−1 for position of anchor holes in Lightning PDU.

      – Drill anchor hole in floor anchor at each mark.

      – Bolt floor anchor to Lightning PDU and to floor.



LOCATE FLOOR ANCHORS
ILLUSTRATION 3−1

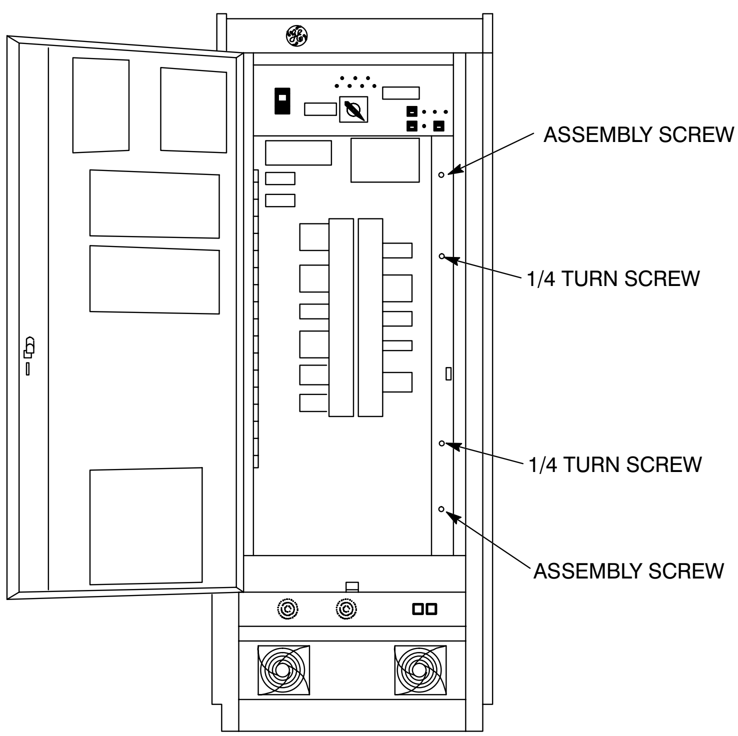
   b. Secure top to wall using wall anchor.

3−1−2 Facility Power and Dedicated Ground Connection

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM LIGHTNING PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Lightning PDU.
3. Measure input voltage test points for 0 VAC before proceeding − if not zero − recheck step 1.
4. Loosen load panel door with a 90˚ turn of two assembly screws on right. See Illustration 3−2.
   - **Fixed site installation.** Remove and discard 1/4−20 screw in center right edge of door.
   - **Mobile installation.** Remove and keep 1/4−20 screw in center right edge of door.



OPEN LOAD PANEL DOOR
ILLUSTRATION 3−2

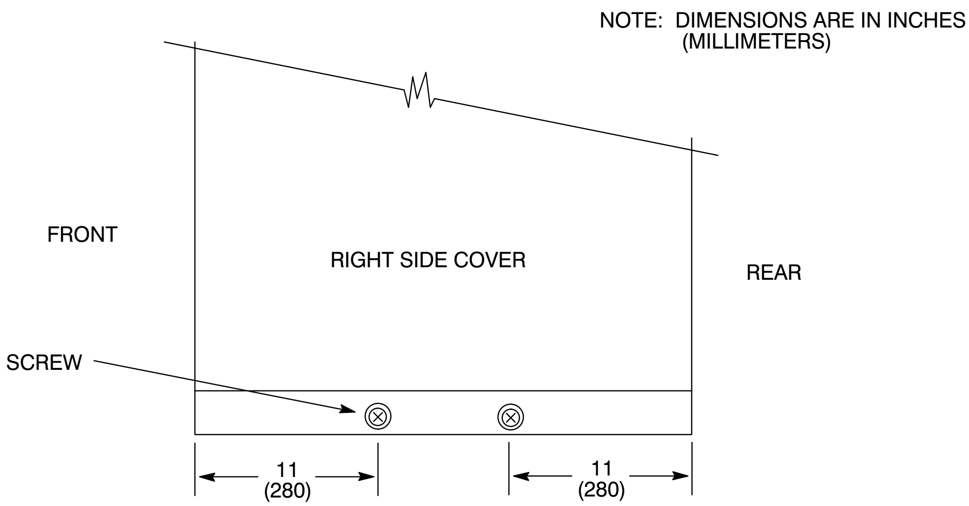
3–1–2 Facility Power and Dedicated Ground Connection (continued)

5. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

6. Remove a side or back panel as follows:

   a. For side panel, remove screw under lower lip. See Illustration 3–3. Grasp both sides of panel and lift up firmly to remove.

   b. For rear panel, remove two screws on each side of panel. Support panel while removing seventh screw, located top center.



REMOVE SIDE PANEL
ILLUSTRATION 3–3

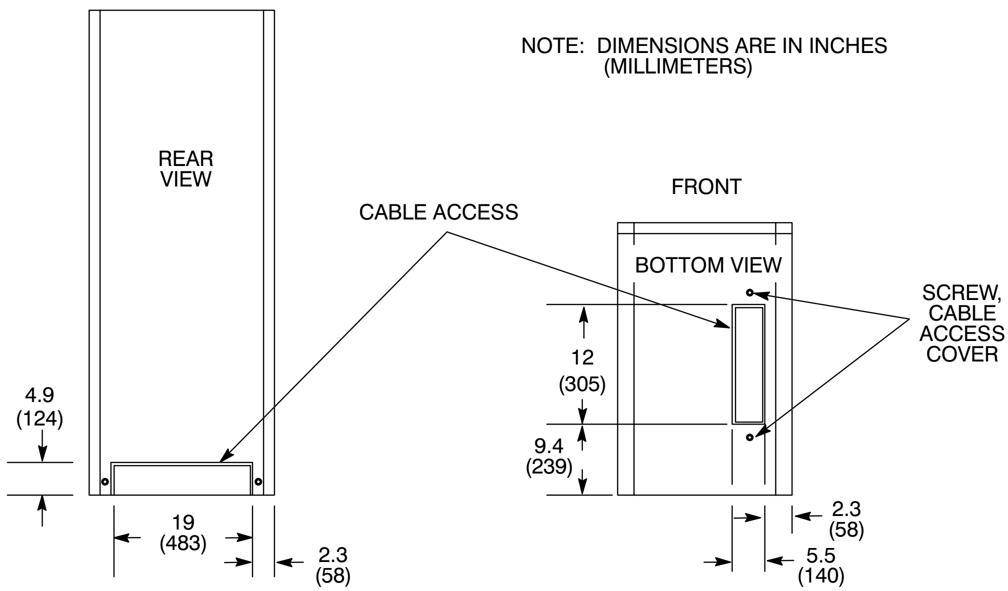
7. Remove a cable access cover. See Illustration 3–4 for cable access location.

   a. Cable access from bottom: Loosen screw on each end of cable access cover. Slide cable access cover to right, then swing up.

   b. Cable access from back: Remove screw on each side of cable access cover. Slide cable access cover down so it clears back cover, and remove.

Cable access cover may be discarded after a permanent installation, or it may be punched or drilled for appropriate conduit fittings.

3−1−2 Facility Power and Dedicated Ground Connection (continued)



CABLE ACCESS
ILLUSTRATION 3−4

8. Route facility input power and ground cables from conduit or cable opening into Lightning PDU. Route to terminal block 10 (TB 10) (A2) and ground bus bar. See Illustration 3−5. Install four conductors to input terminal block on TB 10 (A2) (phase A, B, C, and neutral).

Note

Route all wires away from transformer. Use extra care if routing wires through base of Lightning PDU.

9. Connect facility dedicated ground conductor (green/yellow) to 3/8−16 screw connection attached to ground bus bar on inside back wall of Lightning PDU, below TB 10 (A2). See Illustration 3−5.

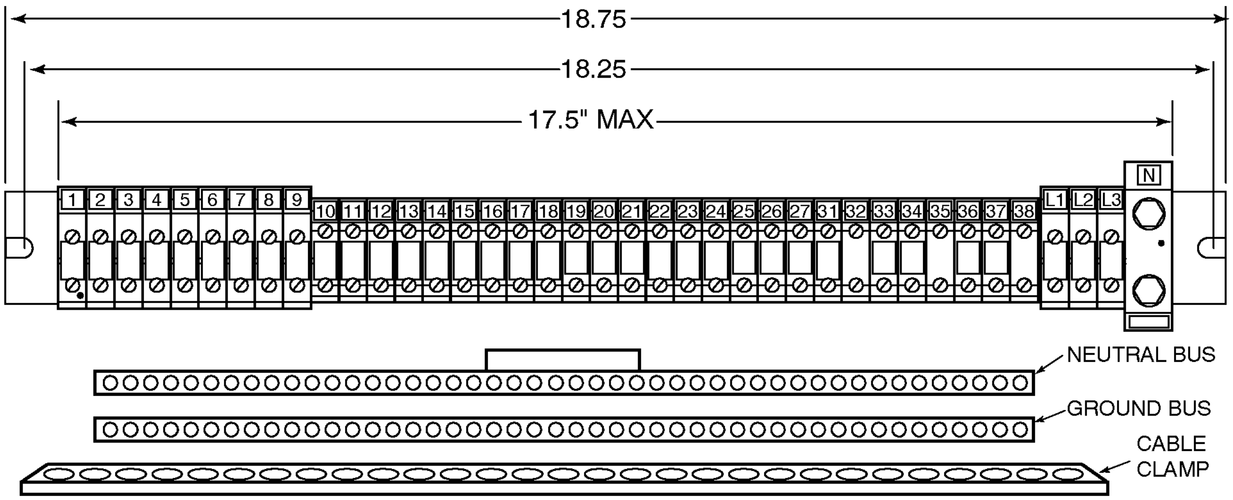
Note

Verify that dedicated ground is same size as incoming power conductors (at least AWG #1/0), or as required by local code.

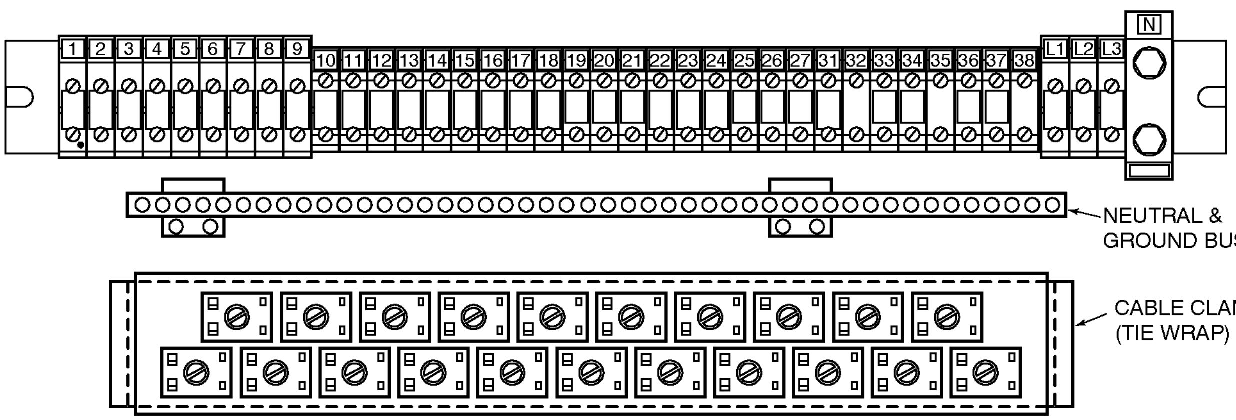
Note

The neutral on TB 10 (A2) is the facility neutral hookup. If facility does not have a neutral LEAVE THIS CONNECTION OPEN. See Illustration 3−5.

3–1–2 Facility Power and Dedicated Ground Connection (continued)



TB10 (A2) WITH CABLE CLAMP CONFIGURATIONS



TB10 (A2) WITH TIE-WRAP CABLE CLAMP CONFIGURATIONS

TB 10 (A2) AND BUS BARS WITH DIFFERENT CABLE CLAMP CONFIGURATIONS
ILLUSTRATION 3–5

10. Inspect all power connections and torque if needed. Refer to Section 3–6–4, PM Procedure, Steps 10 and 16.
11. Check connectors on circuit boards to verify proper seating. Refer to Section 3–6–4, PM Procedure, Step 17.
12. Refer to *Direction 15400, Signa Advantage System*, Set Up and Calibration tab, and perform Section 3, LINE VOLTAGE CHECKS.
13. Verify that incoming power meets requirements. Refer to *Direction 15402, Signa Advantage 1.5T, 1.0T, & 0.5T Site Planning*, Section 5, POWER REQUIREMENTS.

3–1–2 Facility Power and Dedicated Ground Connection (continued)

14. Turn all load panel circuit breakers OFF.

Note

Route all wires away from transformer. Use extra care if routing wires through base of Lightning PDU.

15. See Illustrations 3–5.

- If unit has cable clamp arrangement as shown in illustration 3–5 (top), then loosen cable clamps. Routing output wires away from transformer, bring them into Lightning PDU through rear or a bottom cable access. Thread output wires through cable clamp to TB10 (A2).
- If unit has tie wrap cable clamp arrangement as shown in illustration 3–5 (bottom), then route cables away from transformer bringing them into Lightning PDU through rear or a bottom cable access bringing them across tie wrap cable clamps up to TB10 (A2). Secure cables via tie wraps which are supplied within unit in a plastic bag.

16. Connect output cables to terminals of TB10 (A2). Refer to Illustration 3–5.
17. Connect each cable safety ground wire to bus bar, below TB10 (A2), and neutral wire to bus bar. See Illustration 3–5.
18. Connect equipotential stud on back of Lightning PDU to next cabinet.

CAUTION

After all connections are made and verified correct, tighten each lug securely. This should be checked a second time, preferably by another person. Be sure all lugs are properly connected and securely tightened. High voltage does not forgive wiring errors or loose connections!

19. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
20. Unlock facility circuit breaker and turn ON.
21. Turn main circuit breaker ON.
22. Press Emer. Stop Reset switch.
23. Turn load panel circuit breakers ON.
24. Close outer door.

3–1–2 Facility Power and Dedicated Ground Connection (continued)

CB NO.	LOAD UNITS	TB10 UNIT	AMPERAGE		AMPERAGE	TB10 UNIT	LOAD UNITS	CB NO.
CB6	GRADIENT AMPLIFIER CABINET #1 (208Y/120)		SHUNT		15A	20	HEAT EXCHANGER	CB3
		1	60A		NONE	21	SPARE	CB4
		2	60A			22	SPARE	CB5
		3	60A			23		
CB7	GRADIENT AMPLIFIER CABINET #2 (208Y/120)		SHUNT			24		
		4	60A		30A	7	GRAM CABINET 208Y/120	CB13
		5	60A		30A	8		
		6	60A		30A	9		
CB16	SUPER CON SHIM POWER SUPPLY (208Y/120)	33	30A		SHUNT			
		34	30A		30A	36	MAGNET POWER SUPPLY 208Y/120	CB15
		35	30A		30A	37		
CB17	RF AMPLIFIER/ CABINET (208Y/120)		SHUNT		30A	38		
		10	30A		NONE		OPERATOR WORKSPACE (PC) (120 VAC)	CB8
		11	30A		15A	31		
		12	30A		15A	32	S/OUTL (120 VAC)	CB9
CB1	CUSTOMER OPTIONS	13	15A		15A	26	GRADIENT WATER CHILLER*	CB18
		14	15A		15A	27		
					SHUNT			
CB11	SYSTEM CABINET (208Y/120)	16	30A		15A	19	MFC (120 VAC)	CB2
		17	30A		15A	N/A	PDU (120 VAC)	CB14
		18	30A		20A	25	OPERATOR WORKSTATION (HOST) (120 VAC)	CB12

Shaded area numbers represent TB10 terminal numbers

* Earlier versions of the P20 and P21 used this location (CB18) for optional Resistive Shim power supply.

LOAD CENTER SILK SCREEN (MODELS P20 AND P21)
ILLUSTRATION 3–6

3−1−3 Functional Checks

1. Open outer door of Lightning PDU.
2. Turn main circuit breaker OFF.
3. Turn all load panel circuit breakers OFF.
4. Turn facility circuit breaker ON.
5. Press Emer. Stop Reset Switch.
6. Turn Main Circuit Breaker ON.
7. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4−20 screw in center right edge of door. See Illustration 3−2. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

DANGER!!

FATAL SHOCK HAZARD!! USE EXTREME CAUTION FOR STEPS 8 THROUGH 18. LETHAL VOLTAGES EXIST ON LOAD CENTER BRACKET ASSEMBLY (A14), AND ON BOTH SIDES OF LIGHTNING PDU INTERIOR. LETHAL VOLTAGES EXIST ON THE REGULATION PANEL.

8. Check that voltage at input terminals of TB10 L1, L2, and L3 matches voltage on nameplate located over load panel.
9. Check that input current is phased in clockwise rotation (phase A, B, C). Use a phase meter. Refer to *Direction 15401, Signa Advantage Subsystems*, PDU: Functional Checks tab, and perform Section 1−1, PHASE SEQUENCE CHECK. Refer to Table 3−1 below for phase relationship of input and output voltages.

TABLE 3−1
PHASE SEQUENCE

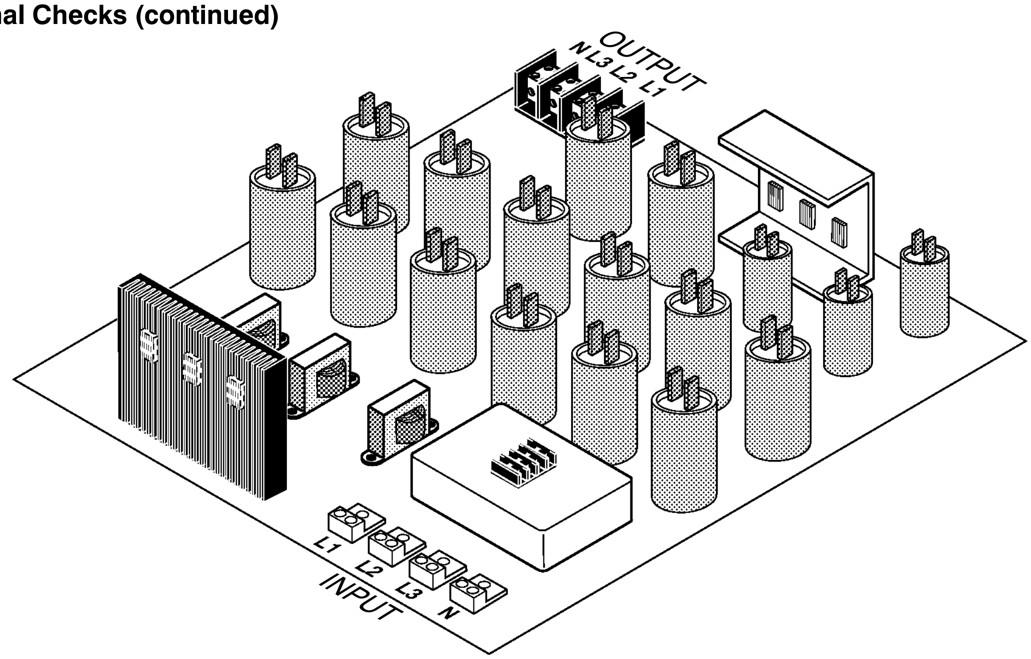
PHASE	1 (L1)	2 (L2)	3 (L3)
PHASE METER	A	B	C
WIRE COLOR	BROWN	RED	ORANGE
H	H_1	H_2	H_3
L	L_1	L_2	L_3
X	X_1	X_2	X_3

CAUTION

Voltage must be phased in clockwise rotation. Improper phase rotation can damage medical equipment.

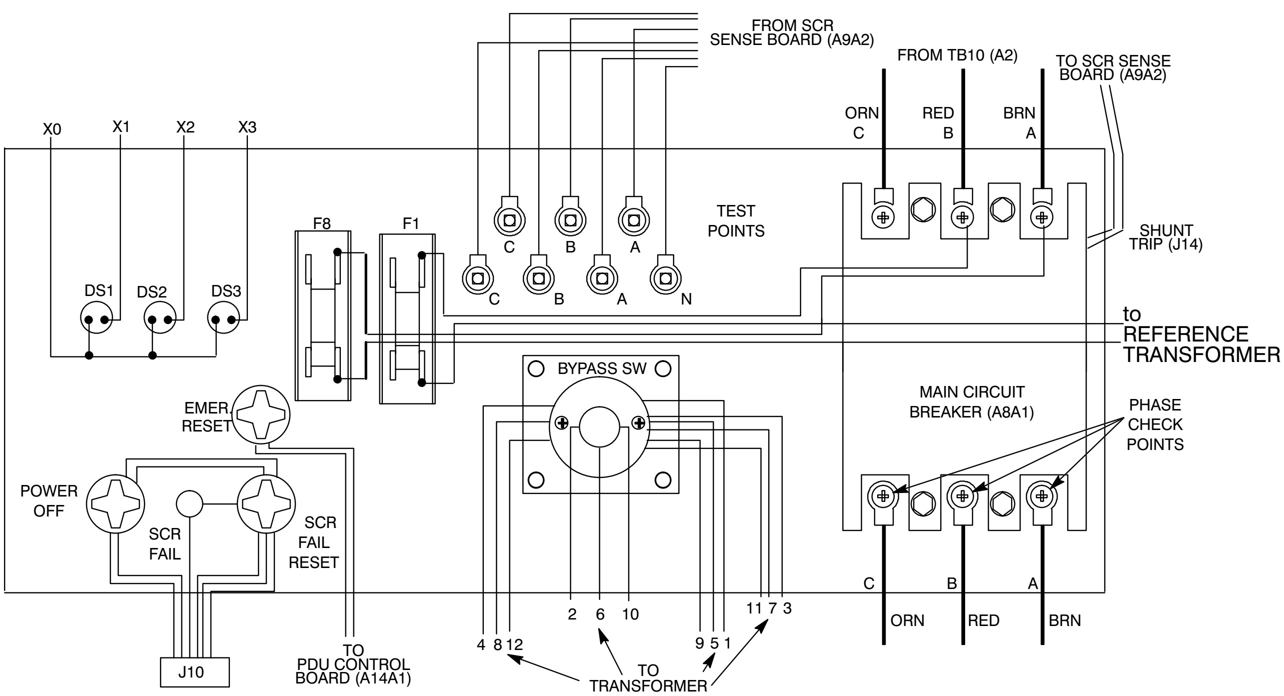
a. Check phasing at primary terminals L1, L2, and L3 line side of filter assembly. See Illustration 3−7.

3–1–3 Functional Checks (continued)



FILTER ASSEMBLY
ILLUSTRATION 3–7

b. Check phasing at bottom of main circuit breaker. See Illustration 3–8.



FRONT PANEL, MODEL 46–317099 P20 AND P21 (REAR VIEW)
ILLUSTRATION 3–8

3–1–3 Functional Checks (continued)

10. Measure facility line–to–line voltage at input test points and compare with model voltage specification given in Table 3–2.

Note

There is a 100:1 voltage stepdown at test points. For example, 480 VAC in Lightning PDU is measured as 4.8 VAC at test points.

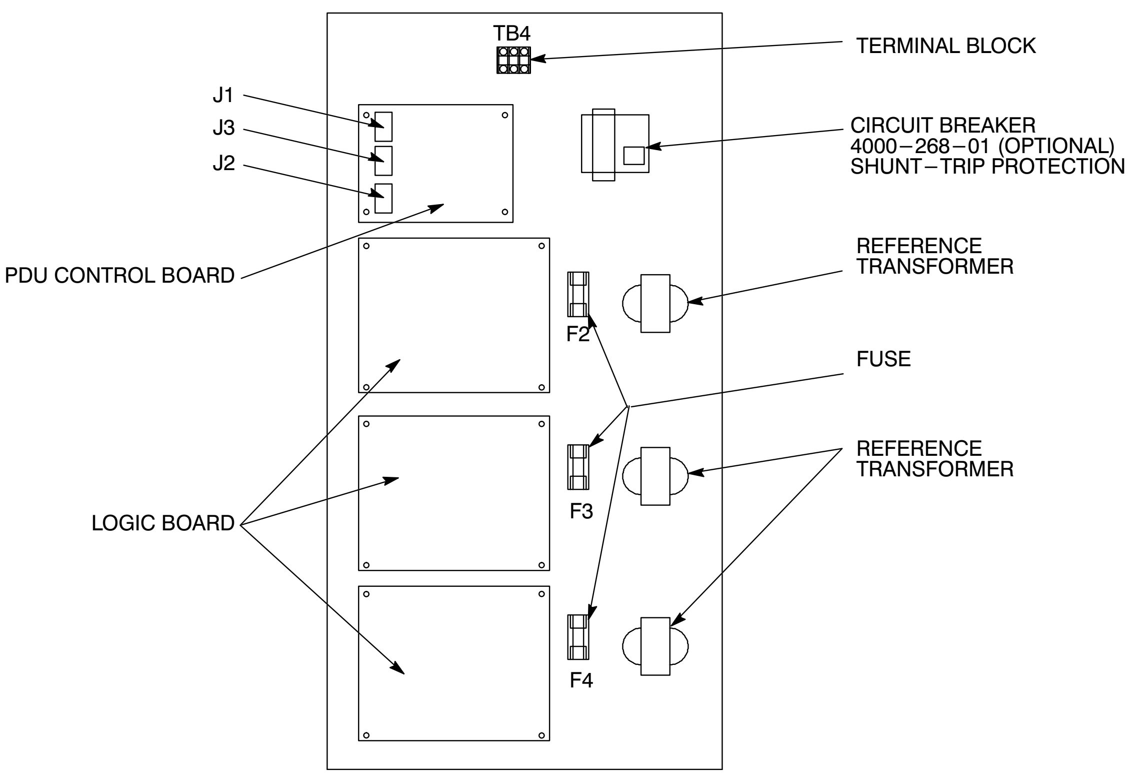
TABLE 3–2
VOLTAGE INPUT BY MODEL

Lightning PDUPART NO.	NOMINAL INPUTLINE–TO–LINE	INPUT VOLTAGE AT INPUT TEST POINTS	INPUT VOLTAGE AT TB10, L1,L2,L3
46–317099P20	480 V 60 Hz regulated, filtered	4.18 to 5.04 V	412 to 502
46–317099P21	380, 400, or 415 V 50/60 Hz regulated, filtered	3.31 to 3.99 V, or 3.48 to 4.20 3.61 to 4.50	331 to 399 348 to 420 361 to 436

Note

Use the next step to perform functional checks without cables connected to J1, J2, and J3 of PDU control board.

11. In documentation package inside outer door, there is a jumpered connector. Insert jumpered connector into mating connector A14A1J2 on PDU control board. See Illustration 3–9.



LOAD CENTER BRACKET ASSEMBLY (A14)
ILLUSTRATION 3–9

3–1–3 Functional Checks (continued)

12. Turn main circuit breaker ON. After 6 seconds, Output indicator lights DS1, DS2 and DS3 light.
13. Press emer. stop reset switch. If indicator lights do not light, refer to Section 3–3, TROUBLESHOOTING.
14. Verify voltage of 208 V 3 phase with neutral at magnet and superconducting shim service outlets. See Illustration 1–4.
15. Verify that the two fans are running. Hold a piece of paper in front of each fan in succession. If paper does not stick to fan guard, check fan. Refer to Section 3–6–4, PM Procedure, Step 19.
16. If voltage is within specification, system is ready to run. If not, refer to Section 3–1–4, Calibrate Logic Board.
17. Turn load panel circuit breakers ON.
18. Verify that voltages on TB10 (A2) match output voltages shown in appropriate Illustration 3–6.
19. Prepare unit for service:
    a. Install any side or rear panels removed.
    b. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
    c. Close outer door.

3–1–4 Calibrate Logic Board

The purpose of this procedure is to set the output of regulator to load.

The driver board has seven green LEDs. Two LEDs should be lighted at all times. LED 7 indicates power on board and is always on. LEDs 1–6 represent tap settings and only one should be on at any time.

If no LED or one LED is lighted, or if more than two LEDs are lighted, immediately turn main circuit breaker OFF. Refer to Section 3–6, TROUBLESHOOTING, for diagnosis and repair.

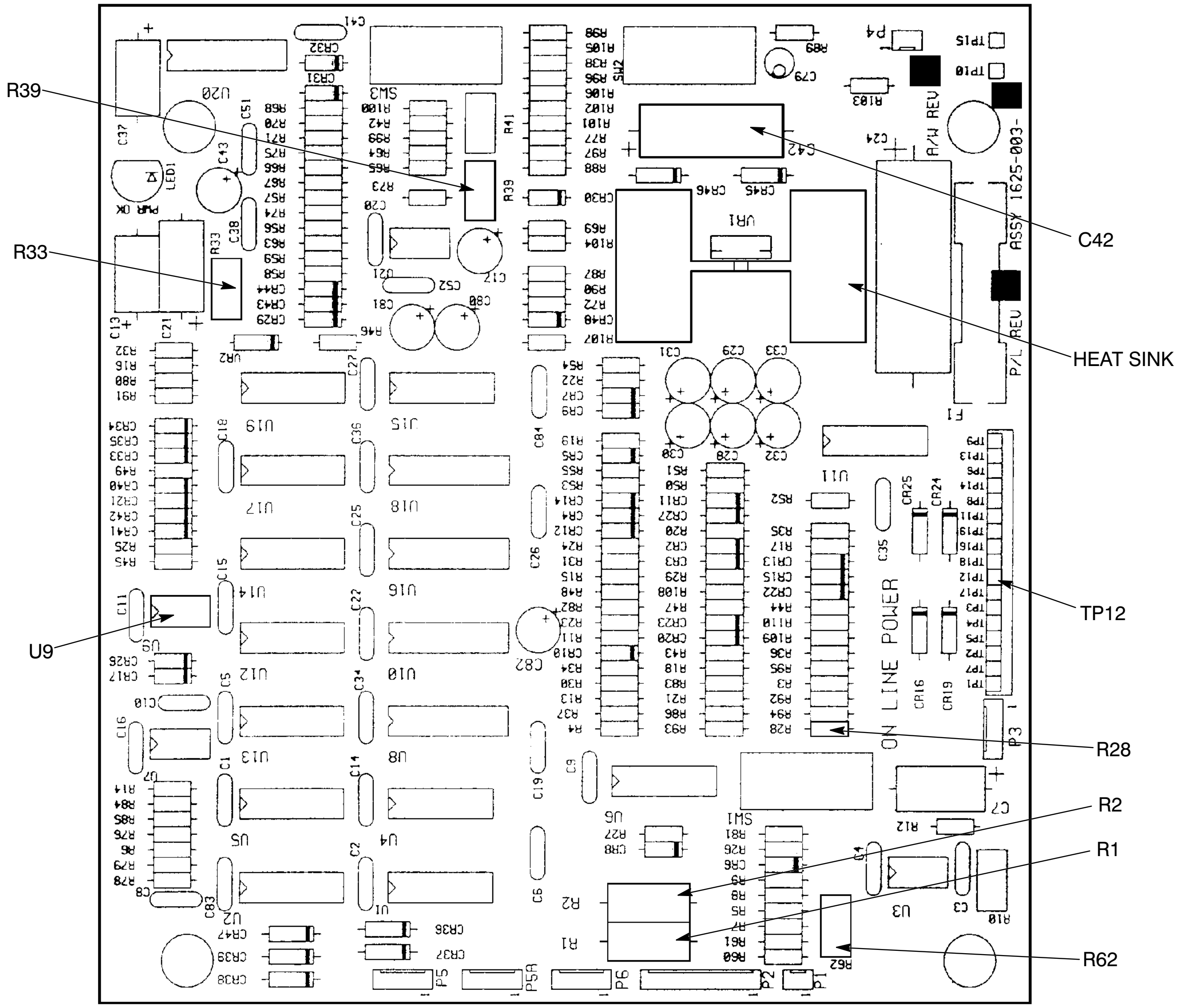
If only one LED (1–6) is lighted, perform the following calibration:

1. Adjust load.
    a. Turn on load and try to achieve 1 KW per phase (8.3 amps)

    Note
    Potentiometer R62 is a 30–turn pot. It may require several turns before output level changes.

    b. Verify voltage across C42 on logic board of 8.5–9.0 VDC. See Illustration 3–10.
    c. If reading is low, adjust R62 to increase reading. If correct reading cannot be obtained, refer to Section 3–3–2, Basic Check, Steps 10–12 to troubleshoot logic board.
    d. Perform step b. for phase B and C.

3−1−4 Calibrate Logic Board (continued)



LOGIC BOARD
ILLUSTRATION 3−10

2. Adjust voltage.

   a. Locate R39 on phase A logic board.

   b. Verify output of phase A to neutral at test point of 1.20 $\pm$ 0.03 V. Measure with a true RMS multimeter.

   Note
   Output voltage does not change linearly as potentiometer R39 is rotated. Output voltage does a step change. R39 is a 30−turn pot. It may require several turns before output level changes.

   c. Adjust R39 to obtain correct output voltage for phase A.

   d. Perform Step 2 for phase B and phase C.

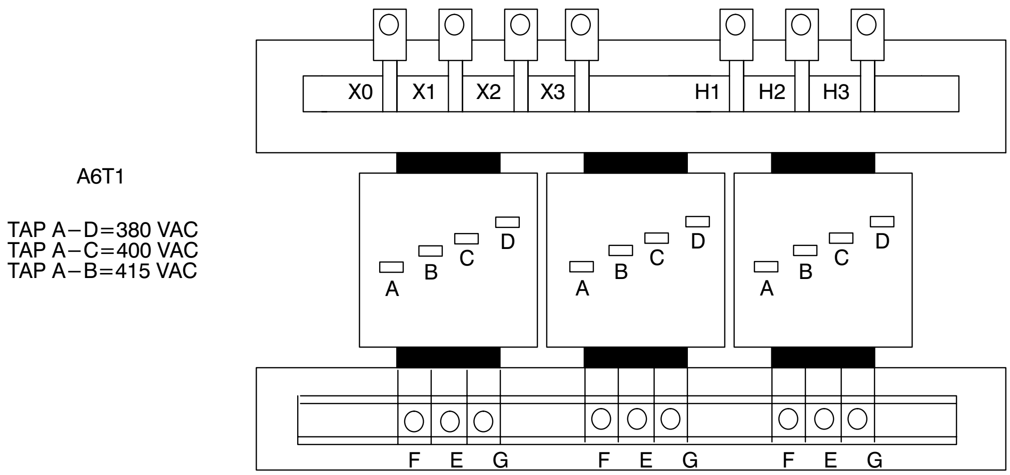
3−2 CONFIGURATION

3−2−1 Model 46−317099P20

Model 46−317099P20 is shipped factory configured. See illustration 3−11 to check or change taps to match input voltage.

3−2−2 Model 46−317099P21

Model 46−317099P21 are shipped factory configured for 380 VAC, 50 HZ input.



TRANSFORMER POWER CONNECTIONS, MODEL 46−317099P21
ILLUSTRATION 3−11

3–3 TROUBLESHOOTING

3–3–1 Prestartup

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK DISCONNECT POWER FROM LIGHTNING PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door to Lightning PDU.
3. Turn main circuit breaker OFF.
4. Measure at power input tests points and reading must be 0 VAC. If not recheck steps 1 and 3.
5. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

Note

The following steps applies to both models 46–317099 P20 and P21. Adjust all three logic boards (A14A2, A14A3, A14A4).

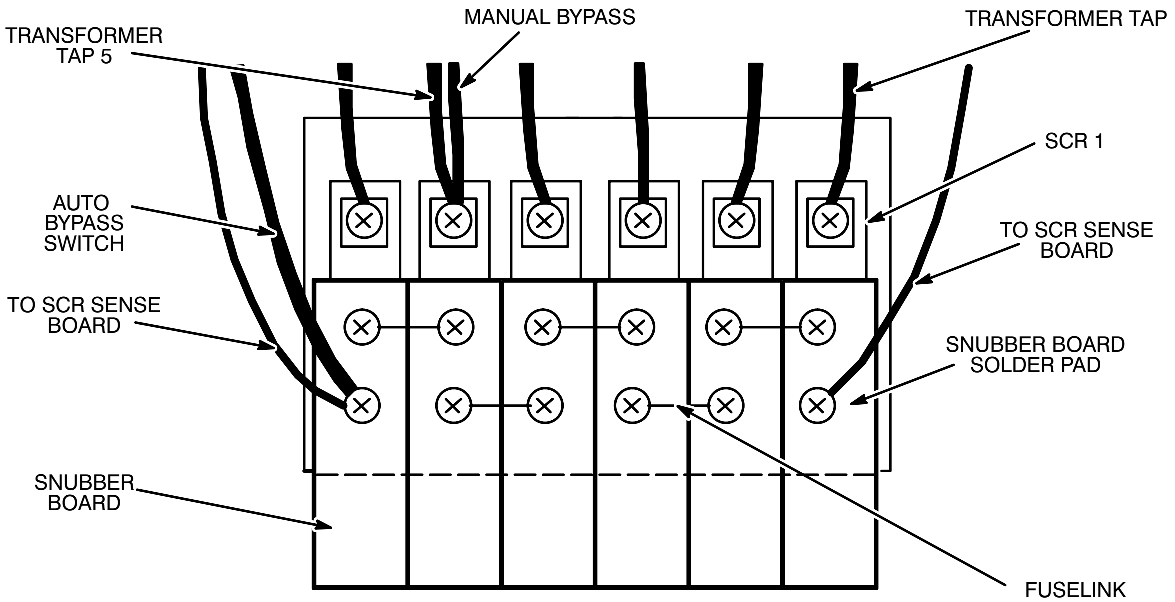
6. On each logic board, attach ohmmeter leads from U9 pin 6 to U9 pin 8, the logic board is oriented as shown in Illustration 3–10. Adjust R33 for ohmmeter to read 105K ohms.
7. Check that all plug connectors on circuit boards are tight. Refer to Section 3–6–4, PM Procedure, Step 15.
8. Tighten screws on all 18 SCRs (54 screws; 3 per SCR). Torque to 30 lb in (34 Nm). See Illustration 3–12.
9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
10. Unlock facility circuit breaker and turn ON.

Note

There is a 100:1 voltage stepdown at test points. For example, 480 VAC in Lightning PDU is measured as 4.8 VAC at test points.

11. Measure facility line–to–line voltage at input test points.

3–3–1 Prestartup (continued)



SCR BANK
ILLUSTRATION 3–12

3–3–2 Basic Check

1. Unlock facility circuit breaker and turn ON.
2. Turn bypass switch to AUTO position.
3. Turn main circuit breaker ON. Press emer. stop reset switch.

Note
After turning main circuit breaker ON there is a 6–second built–in delay before power is applied to SCRs. Detect an SCR failure by measuring neutral–to–line output voltage of each phase. Test point voltages are in the range 0–5 VAC.

4. Set meter to 2 VAC scale.

Note
Measured voltage should be close to zero. If measured voltage is greater than 0.2 VAC, there is a shorted SCR in regulation panel. Refer to Section 3–3–4, SCR troubleshooting.

a. Observe meter after 6 seconds. Verify test point A line–to–neutral output voltage of 1.20 $\pm$ 0.03 V.

3−3−2 Basic Check (continued)

b. Press power off switch.

c. Turn main circuit breaker OFF then ON to reset.

d. Move meter lead to output test point B. Perform Steps a. through c.

e. Move meter lead to output test point C. Perform Steps a. through c.

5. Turn main circuit breaker ON.

6. Press Emer. Stop Reset switch.

7. Turn load panel circuit breakers ON.

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN LIGHTNING PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

8. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4−20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

9. Check each set of output line−to−neutral voltages at TB10 (A2). Refer to Illustration 3−6 and 3−11

10. Adjust logic board for phase A as follows. See Illustration 3−10.

a. Measure AC voltage across resistor R1 or R2. Verify that voltage is between 10−100mv.

Note

If voltage is zero, (1) there is no load, or (2) current sensor is defective or disconnected. Logic board does not regulate if voltage is zero.

b. Measure DC voltage across C42. Adjust R62 so voltage measures 8.5−9.0 VDC.

c. Adjust R39 counterclockwise so that LED 6 on driver board is lit.

d. Measure DC voltage across C42. Adjust R62 so voltage measures 8.5−9.0 VDC.

e. Monitor neutral−to−phase−\a output voltage. Adjust R39 so output line−to−neutral voltage is 1.2±0.03V.

11. Adjust logic boards for phase B and phase C. Refer to Step 10.

3−3−2 Basic Check (continued)

12. Verify that voltage at TP12 to ground on each logic board is 10 $\pm$ 0.5 VDC. If voltage is not within these limits, replace logic board.

13. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4−20 screw in center of right edge of door.

14. Close outer door to Lightning PDU.

3−3−3 Functional Check

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK DISCONNECT POWER FROM LIGHTNING PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock, and tag.

2. Check and replace any fuselinks on the SCR sub−assemblies which are discolored or which are not continuous.

3. See Illustration 3−12. For each SCR bank:

   Note
   SCRs are numbered from right to left. SCR number six is on the left when SCRs are oriented as shown in Illustration 3−12.

   a. Turn bypass switch to the manual position.

   b. Verify that resistance between an SCR transformer lead and a snubber board solder pad exceeds 1K ohms. If not, refer to Section 3−3−4, SCR Troubleshooting.

4. Check snubber boards for discoloration and rupture of foil.

5. Check that all snubber board plug connections are tight.

6. Check that logic, driver, and SCR sense boards are firmly supported on standoffs.

7. Check that all plug connectors on circuit boards are tight. Refer to Section 3−6−4, PM Procedure, Step 15.

8. Check for heat discoloration near heat sink area on logic boards. See Illustration 3−10.

9. Check that all connectors to input/output filter assembly are tight (A11).

3–3–3 Functional Check (continued)

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN LIGHTNING PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

10. Unlock facility circuit breaker and turn ON.
11. Turn main circuit breaker ON.
12. Press emer. stop reset switch.
13. Press power off switch. This shunt trips main circuit breaker.
14. Measure input test points – must measure 0 VAC before proceeding.
15. Remove fuselink from any one SCR bank.
16. Turn main circuit breaker OFF then ON to reset. Main circuit breaker will trip in about 10 seconds. If it does not trip, replace SCR sense board.
17. Install fuselink. Turn main circuit breaker OFF then ON to reset.
18. Press emer. stop reset switch.
19. Check output line–to–neutral voltage at test points (A8). Verify voltage of 1.20 $\pm$ 0.03 V. If voltage is out of specification, refer to Section 3–3–2, Basic Check, for logic board adjustment.
20. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
21. Turn load panel circuit breakers ON.
22. Close outer door.

3–3–4 SCR Troubleshooting

1. Turn main circuit breaker OFF.

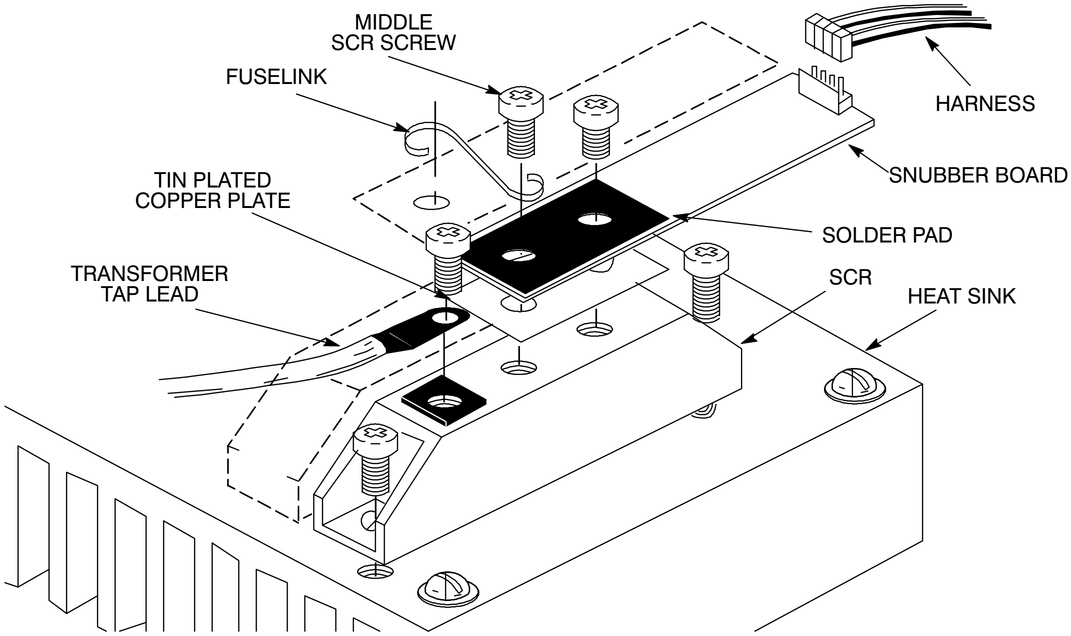
Note

SCRs are numbered from right to left. SCR number six is on the left when SCRs are oriented as shown in Illustration 3–12.

2. Turn bypass switch to the manual position.

3. For SCR number one:

   a. Measure resistance between transformer tap connection and solder pad area of snubber board. If reading is greater than 1000 ohms, proceed to Step 5. If reading is less than 1000 ohms, continue with this step. Either SCR, snubber board, or driver board is defective. See Illustration 3–13.



SCR ASSEMBLY
ILLUSTRATION 3–13

   b. Disconnect harness between driver board and snubber board. Measure resistance between transformer tap lead and each middle SCR screw.

   c. If reading is greater than 1000 ohms then driver board is defective. Replace driver board. Proceed to Step 4f. If reading is less than 1000 ohms then either SCR or snubber board is defective. Continue with step 4d.

   d. Disconnect snubber board from SCR. Measure resistance between tap connection and screw in middle of SCR. If resistance is less than 1000 ohms, replace SCR. If resistance is greater than 1000 ohms, replace snubber board.

3–3–4 SCR Troubleshooting (continued)

e. Reconnect snubber board to SCR.

f. Reconnect harness between driver board and snubber board.

4. Repeat Step 4 for each remaining SCR in the SCR bank.

5. Reconnect fuselinks between snubber boards. Replace any fuselink which is open or discolored.

6. Replace any snubber board which has cracked or discolored foil pads on either side of board.

7. Repeat Steps 2–8 for each remaining SCR bank.

8. Turn bypass switch to AUTO position.

9. Measure at output test points (A8) to see if a shorted SCR is doing an instant output voltage.

Note
Steps 11a, 11b and 11c must be finished within six seconds.

a. Turn main circuit breaker ON.

b. Measure voltage across phase A to neutral at output test points before 6 seconds has elasped.
If measurement is between 1 to 2 VAC in the 6 second period, go to Step 10.
If no voltage in the first 6 seconds, the phase is O.K. and test phase B and C.

c. Turn main circuit breaker OFF.

10. SCR or snubber board is defective. Perform this step to determine which part is defective.

a. Disconnect snubber board from SCR.

b. Measure resistance between tap connection and screw in middle of SCR.

c. If resistance is less than 1000 ohms, replace SCR. Refer to Section 3–4–2, Replace an SCR.
If resistance is $\geq$ 1000 ohms, replace snubber board. Refer to Section 3–4–1, Service a Circuit Board.

d. Continue with Step 9.

11. Turn main circuit breaker ON.

3–3–5 Filter Troubleshooting

1. Open outer door of Lightning PDU.

2. Turn main circuit breaker OFF.

3. Turn all load panel circuit breakers OFF.

3–3–5 Filter Troubleshooting (continued)

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN LIGHTNING PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

4. Loosen load panel door with a 90˚ turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

5. Turn main circuit breaker ON. If main circuit breaker shunt trips, check Accuvar "O.K." light is ON. If fail light is ON replace Accuvar.

   a. Verify that the green OK light is ON.

   Note

   Voltage measured at the primary terminal will be nominal input volts to the PDU. These are the terminal lugs in the front of the filter assembly

   b. Verify that line–to–line voltage on secondary side of filter terminal block at rear of the filter assembly is in operating voltage range (208Y/120). See illustration 3–7.

6. Check for visible damage to filter.

   a. If a terminal block is damaged, replace terminal block.

   b. If surge suppressor is damaged, this will be indicated by the red fail light ON located on the Accuvar. Replace Accuvar. Refer to Section 3–4–5, Replace Surge Suppressor.

   c. If any capacitors are damaged, replace the capacitors.

7. If there is no visible damage, disconnect wires from line and load side of surge suppressor.

8. Turn main circuit breaker ON.

   a. If main circuit breaker shunt trips, the problem is in the PDU

   b. Turn main circuit breaker OFF.

   c. Replace the primary power wires to the line filter.

   d. Turn main circuit breaker ON. If the main circuit breaker does not trip, the primary filter is OK. If the main circuit breaker does trip, replace the filter.

   e. If the primary filter is OK, repeat steps C and D above for the secondary filter. If primary or secondary filter causes circuit breaker to trip replace filter.

3−3−5 Filter Troubleshooting (continued)

9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4−20 screw in center of right edge of door.

10. Turn main circuit breaker ON.

11. Turn load panel circuit breakers ON.

12. Close outer door.

3−3−6 Miscellaneous Problems

PROBLEM: Main circuit breaker fails to shunt trip.

PROCEDURE:

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN LIGHTNING PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

1. Disconnect connector on left side of main circuit breaker.

2. Measure shunt trip coil resistance with circuit breaker turned ON. It must read <10 ohms. With circuit breaker OFF, the shunt trip coil must be open. If either condition is not met, replace circuit breaker.

3. Reconnect connector behind main circuit breaker.

4. Turn main circuit breaker ON.

5. Measure shunt trip coil voltage on cable side of unplugged connector J1 with power off switch pressed. If voltage is 24−28 VDC, replace main circuit breaker. Refer to Section 3−4−3, Replace Main Circuit Breaker.

6. If no voltage, check fuses F1 and F8 on A8. If fuse failed, replace fuse.

PROBLEM: Main circuit breaker shunt trips on powerup with bypass switch in AUTO position.

PROCEDURE:

1. If SCR failure is lighted, replace any open fuselinks, defective snubber boards, or shorted SCRs. Refer to Section 3−3−4. Press fail reset switch to clear SCR failure light.

2. Turn main circuit breaker OFF. Disconnect SCR sense board J9. Turn main circuit breaker ON. If breaker does not shunt trip, replace SCR sense board. Refer to Section 3−4−1.

3−3−6 Miscellaneous Problems (continued)

PROBLEM: No LED on driver board is lit, except LED 7 (Power OK), with bypass switch in AUTO position.

PROCEDURE:

1. Check connection between logic board and driver board. Connector is P2 on logic PCB and P1 on driver PCB.
2. Check input power for clockwise phase rotation Refer to Section 1−4−2.
3. Measure voltage across SCR between transformer tap lead and snubber solder pad. When SCR is conducting, measured voltage should be less than 1.2 VAC.
4. If no SCRs are conducting, replace driver board. Refer to Section 3−4−1, Service a Circuit Board.
5. If SCRs are still not conducting, replace logic board. Refer to Section 3−4−1, Service a Circuit Board.
6. Check voltage across R1 or R2 on logic board. Voltage is typically 10 − 100 mVAC. If it is zero, current signal is disabled from logic board. Check current sense transformer connections. Repair any opens.

PROBLEM: All LEDs on driver board are off.

PROCEDURE:

1. Check interconnect cables to driver board.
2. Check LED 1 on logic board (Power OK). If LED 1 is not lighted, replace logic board. Refer to Section 3−4−1, service a circuit board.

PROBLEM: No voltage output in AUTO mode.

PROCEDURE:

1. Check that main input breaker is ON.
2. Check that load panel circuit breakers are ON.
3. Check that facility circuit breaker is ON.
4. Check that line input power connections are made to H_1, H_2, and H_3 transformer terminal lugs.

PROBLEM: Fuselink opens during powerup.

PROCEDURE:

1. Check that primary RC filter is installed and connected.
2. Check that capacitor connections are tight.
3. Check adjustment for R33 on all 3 logic PCB’s. Refer to Section 3−3−1, Prestartup.

3–3–6 Miscellaneous Problems (continued)

PROBLEM: Main circuit breaker trips.

PROCEDURE:

1. Perform prestartup checks. Refer to Section 3–3–1, Prestartup.
2. If problem persists, replace driver and logic boards. Refer to Section 3–4–1, Service a Circuit Board.

PROBLEM: Lightning PDU does not regulate.

PROCEDURE:

1. Perform prestartup. Refer to Section 3–3–1, Prestartup.
2. Perform basic check. Refer to Section 3–3–2, Basic Check.
3. If problem persists, replace driver and logic boards. Refer to Sections 3–4–1, Service a Circuit Board.

PROBLEM: High line–to–neutral output voltage measured within six seconds after powerup.

PROCEDURE: There is a shorted SCR. Refer to Section 3–3–4, SCR Troubleshooting.

PROBLEM: Regulation panel is running hot.

PROCEDURE:

1. Check for loose power connections.
2. Check that fans are operating.
3. Check for loose connections between driver boards and snubber boards.
4. Check for loose connections between snubber board and SCR.
5. Replace snubber board for conducting SCR if problem persists. Refer to Section 3–4–1, Service a Circuit Board.

PROBLEM: Audible noise comes from regulation panel.

PROCEDURE:

1. Verify that voltage across conducting SCR is less than 1.2 VAC.
2. Perform prestartup. Refer to Section 3–3–1, Prestartup.
3. Perform basic check. Refer to Section 3–3–2, Basic Check.
4. Replace driver boards if problem persists. Refer to Section 3–4–1, Service a Circuit Board.

3−3−6 Miscellaneous Problems (continued)

PROBLEM: Gradient Amplifier, RF Amplifier, and Gram Cabinet circuit breakers trip without a power off actuation.

PROCEDURE:

1. Verify that input voltage is present. Facility input power may have failed.
2. Verify main circuit breaker is ON.
3. Press emer. stop reset, and manually reset circuit breakers.

3–4 LIGHTNING PDU SERVICE

3–4–1 Service a Circuit Board

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

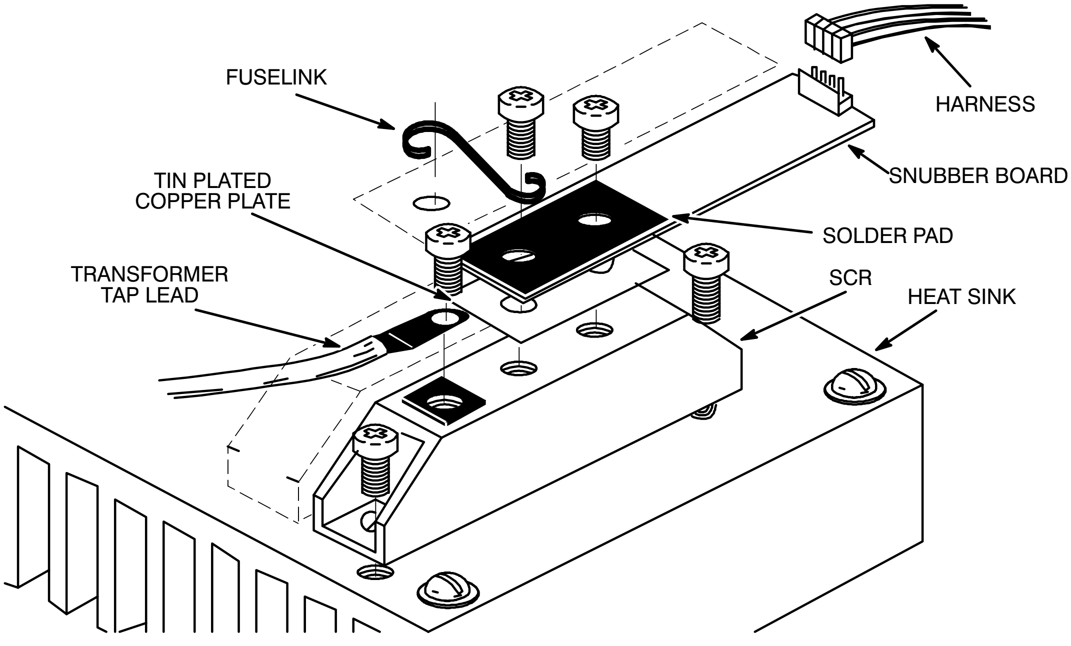
1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Lightning PDU.
3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Turn main circuit breaker OFF.
5. Turn load panel circuit breakers OFF.
6. Measure input test points (A8) for 0 VAC before proceeding.
7. Disconnect connectors on circuit board. Remove board from standoffs.
8. Install replacement board on standoffs. Reconnect connectors.
9. If a logic board is replaced, perform prestartup and basic check on logic board. Refer to Sections 3–3–1, Prestartup, and 3–3–2, Basic Check.
10. Perform functional check. Refer to Section 3–3–3, Functional Check.
11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
12. Unlock facility circuit breaker and turn ON.
13. Turn main circuit breaker ON.
14. Press Emer. Stop Reset switch.
15. Turn load panel circuit breakers ON.
16. Close outer door.

3−4−2 Replace an SCR

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Lightning PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90˚ turn of two assembly screws on right. Mobile installation: Remove and keep 1/4−20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Disconnect transformer tap lead from SCR. See Illustration 3−14.



SCR ASSEMBLY
ILLUSTRATION 3−14

8. Remove fuselinks and snubber board from SCR. Remove SCR from heat sink.

3–4–2 Replace an SCR (continued)

9. Spread a thin film of heat–sink compound on new SCR and mount on heat sink.

10. Attach snubber board to SCR with copper plate under the snubber PCB. Attach fuselinks to snubber board. Torque SCR screws to 30 lb in (34 Nm).

11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center of right edge of door.

12. Perform basic check. Refer to Section 3–3–2, Basic Check.

13. Perform functional check. Refer to Section 3–3–3, Functional Check.

14. Close outer door.

3–4–3 Replace Main Circuit Breaker (A8A1)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.

2. Open outer door of Lightning PDU.

3. Turn main circuit breaker OFF.

4. Measure input test points (A8) for 0 VAC before proceeding.

5. Turn all load panel circuit breakers OFF.

6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

7. Remove main circuit breaker.

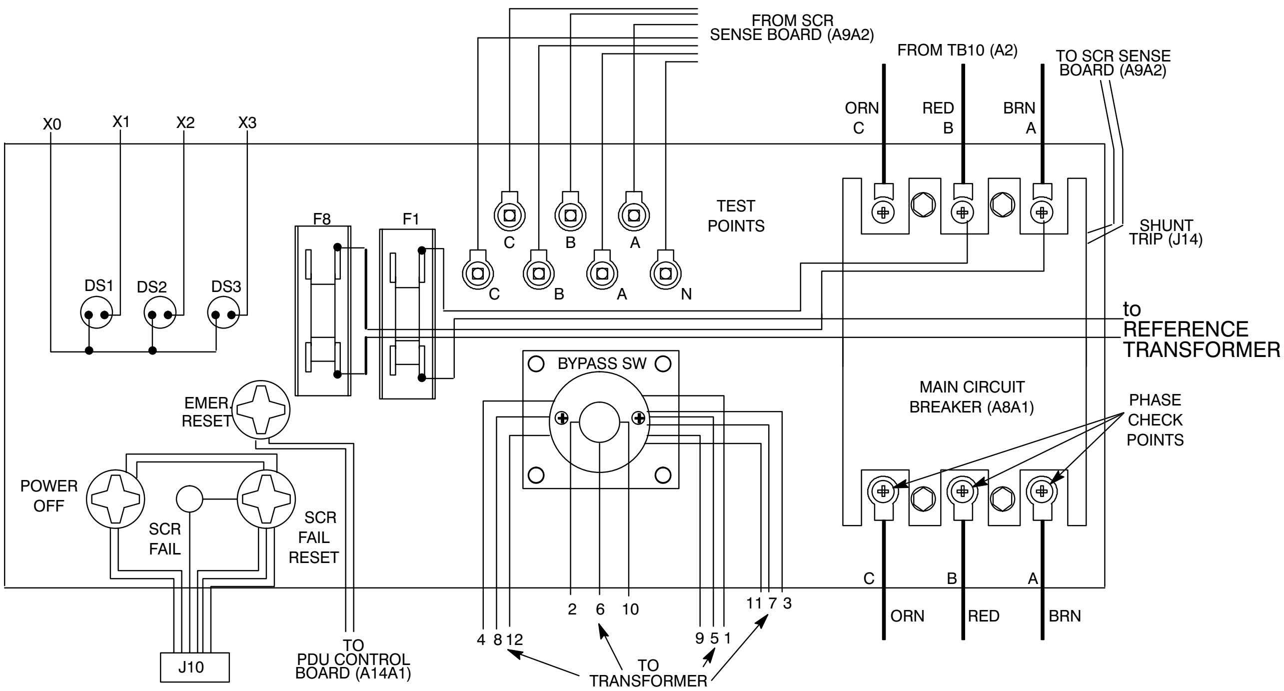
   a. Disconnect main circuit breaker shunt trip (J14). See Illustration 3–15.

   b. Remove two small wires at top of main circuit breaker.

   c. From front panel side, remove and keep four screws, eight washers and four nuts securing main circuit breaker to front panel and 2 spacer plates.

   d. Remove six power connections to main circuit breaker. Remove main circuit breaker.

3–4–3 Replace Main Circuit Breaker (A8A1) (continued)



FRONT PANEL (REAR VIEW)
ILLUSTRATION 3–15

8. Replace main circuit breaker.

Note
Be sure wire color on input is the same as wire color on output!

   a. Install six power connections to main circuit breaker. Be sure ring terminals are in place on top red and brown power connections. Torque to 25 lb in (113 Nm).

   b. Holding circuit breaker in place, install four screws, eight washers and four nuts securing main circuit breaker to front panel. Use the two spacer plates.

   c. Install two wires at top of main circuit breaker.

   d. Connect main circuit breaker shunt trip (J14).

9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.

10. Unlock facility circuit breaker and turn ON.

11. Turn main circuit breaker ON.

3–4–3 Replace Main Circuit Breaker (A8A1) (continued)

12. Press emer. stop reset switch.
13. Turn load panel circuit breakers ON.
14. Close outer door.

3–4–4 Replace Bypass Switch (A8A2)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Lightning PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove bypass switch. See illustration 3–15.
   a. On front panel, remove and keep screw securing knob to bypass switch. Remove knob.
   b. Remove and keep four screws securing bypass switch to front panel. Bypass switch will come free of panel.
   c. Tag wires on bypass switch to identify where they go on the new bypass switch.
   d. Remove wires from bypass switch.
   e. Remove bypass switch.
8. Replace bypass switch.

CAUTION

Connect wires to the same location on bypass switch from which they were disconnected. Severe damage to equipment may result if wiring is incorrect.

   a. Connect wires to bypass switch. Torque nine power connections to 30 lb in (34 Nm).

3–4–4 Replace Bypass Switch (A8A2) (continued)

   b. Holding bypass switch in place, install four screws securing bypass switch to front panel.

   c. Holding knob in place, install screw securing knob to bypass switch.

9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.

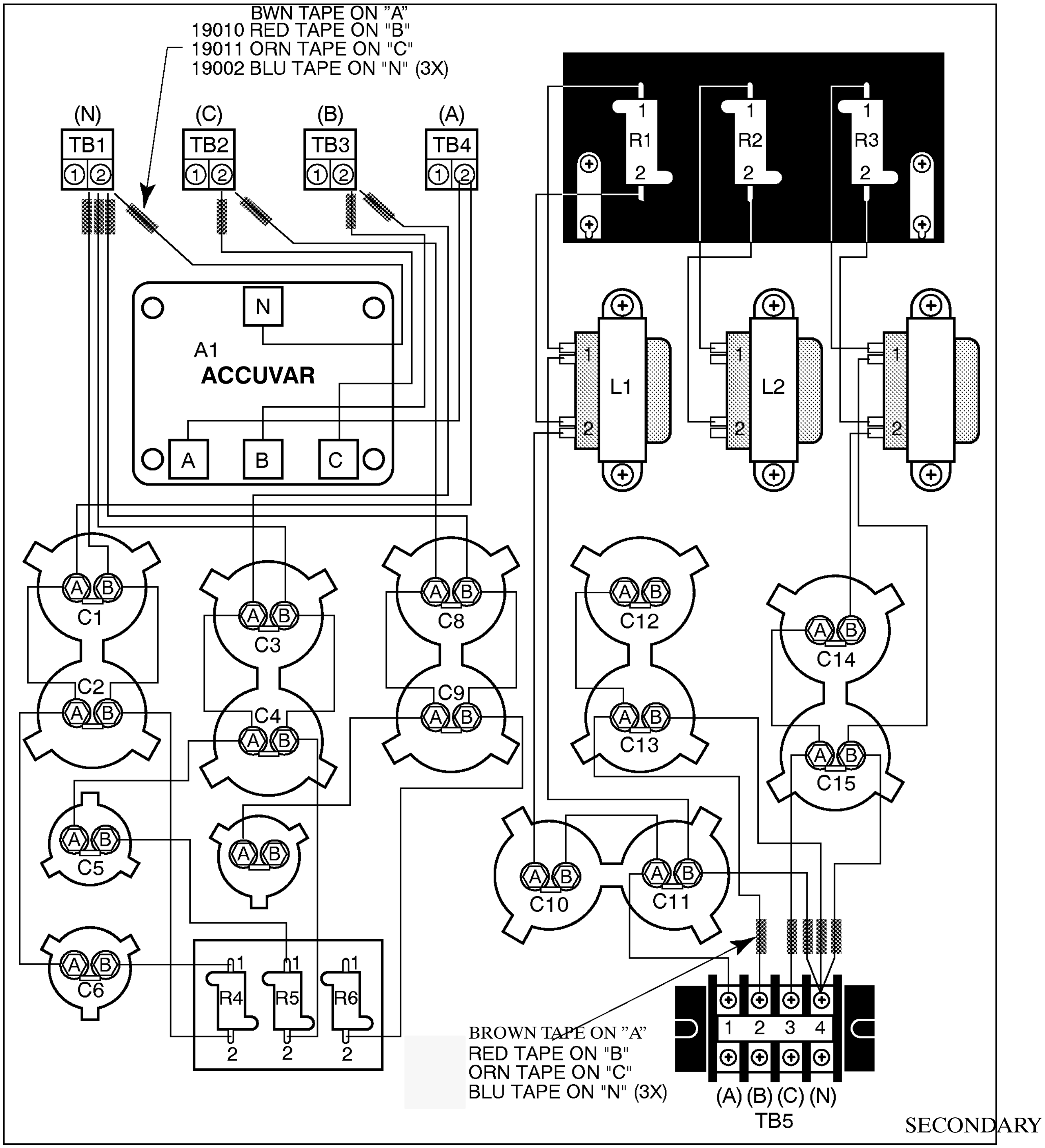
10. Unlock facility circuit breaker and turn ON.

11. Turn main circuit breaker ON.

12. Press Emer. Stop Reset switch.

13. Turn load panel circuit breakers ON.

14. Close outer door.



FILTER WIRING DIAGRAM
ILLUSTRATION 3–16

3–4–5 Replace Surge Suppressor (A11A1)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Lightning PDU.
3. Turn off main input circuit breaker.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
6. Remove and keep four nuts, phillips screws, star washers and flat washers securing surge suppressor mount to filter assembly. See Illustration 3–16.
7. Remove surge suppressor assembly.
8. Install new surge suppressor assembly. Secure to filter assembly with screws, star washers and flat washers kept in Step 6.

Note

Be sure wires get to the correct locations: brown to phase A, red to phase B, orange to phase C, and white or blue to neutral on both line and load sides. See Illustration 3–16.

9. Connect all output wires from surge suppressor.
10. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
11. Unlock facility circuit breaker and turn ON.
12. Turn main circuit breaker ON.
13. Press Emer. Stop Reset switch.
14. Turn load panel circuit breakers ON.
15. Close outer door.

3–4–6 Change Transformer Tap–lead Position (A6T1)

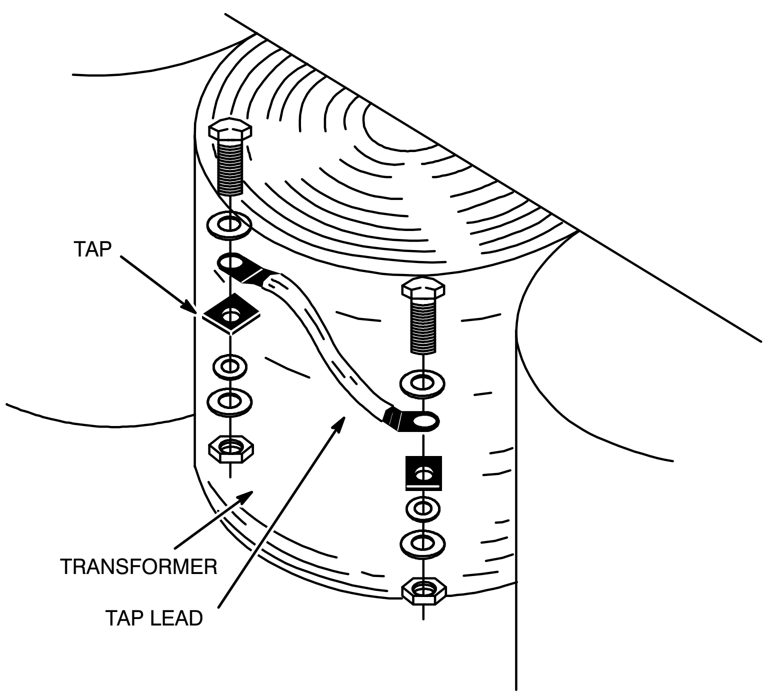
DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Lightning PDU.
3. Measure input test points (A8) for 0 VAC before proceeding.
4. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
5. Unbolt end of tap lead. See Illustration 3–17.
6. Move tap lead to desired tap. Refer to Section 3–2, CONFIGURATION.

Note

There must be a flat washer on each side of a terminal lug. This is a UL requirement.



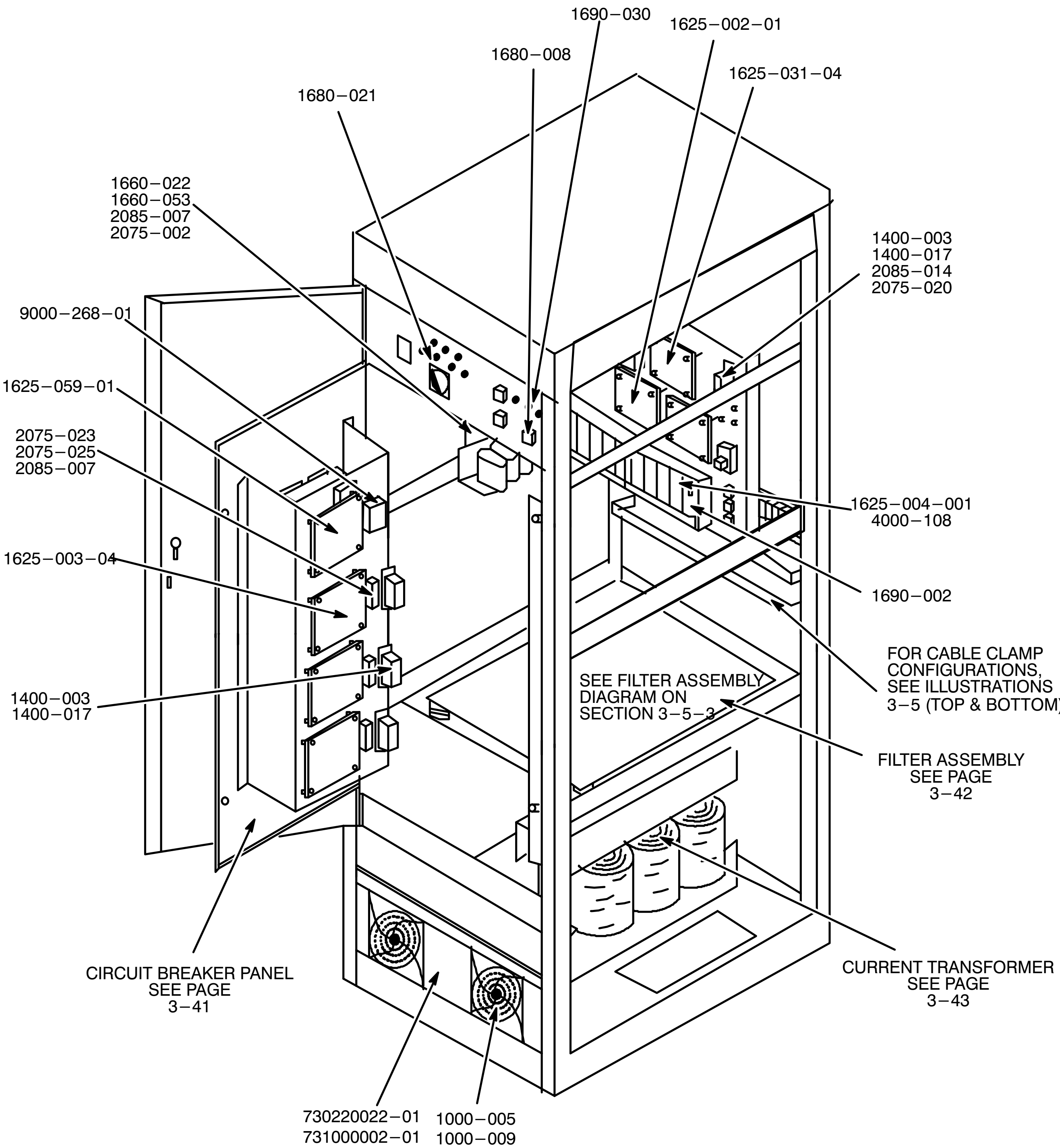
TRANSFORMER TAP CONNECTION
ILLUSTRATION 3–17

3−4−6 Change Transformer Tap−lead Position (A6T1) (continued)

7. Bolt tap lead. Torque to 60 lb in (68 Nm).
8. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4−20 screw in center right edge of door.
9. Unlock facility circuit breaker and turn ON.
10. Press Emer. Stop Reset switch.
11. Turn main circuit breaker ON.
12. Turn load panel circuit breakers ON.
13. Close outer door.

3–5 RENEWAL PARTS

3–5–1 VENDOR PARTS LIST USED IN LIGHTNING PDU

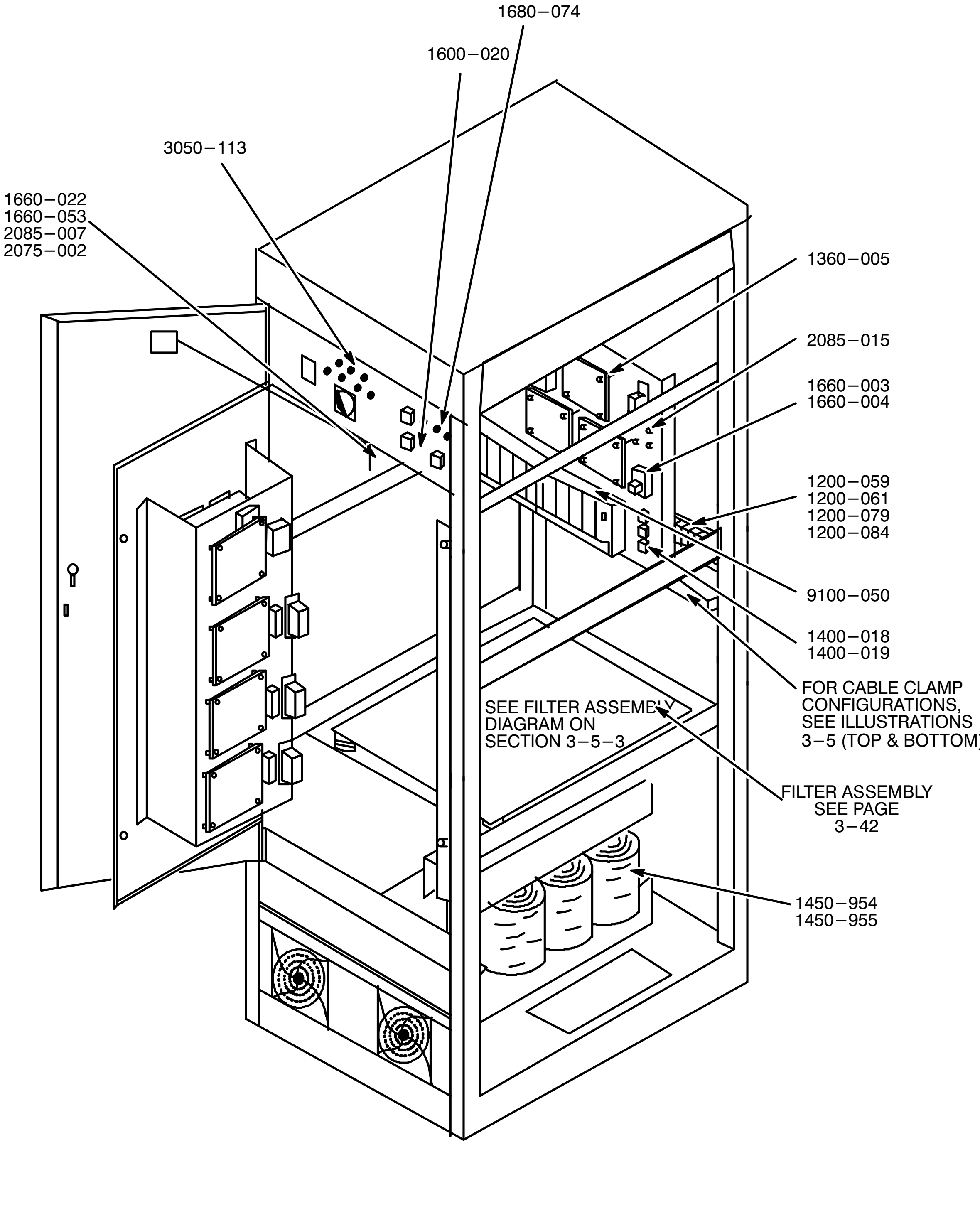


LIGHTNING PDU
ILLUSTRATION 3–18

3−5−1 VENDOR PARTS LIST USED IN LIGHTNING PDU (continued)

Vendor No.	GE Part NO.	FRU	Vendor	Description (Remarks)
7025−020			On−Line Power	FILTER ASSEMBLY, USED ON P20 AND P21
1425−003			On−Line Power	TRANSFORMER, CURRENT, 3 USED, USED ON P20 AND P21
1000−005	46−307819P37	1	On−Line Power	FAN, 115VAC 115 CFM, 2 USED
1000−009	46−307819P76	2	On−Line Power	FILTER, FAN, 4.5 in. MUFFIN, 2 USED
1400−003	46−307819P3	1	On−Line Power	TRANSFORMER, REFERENCE 480 VIN 40 VCT OUT 40 VA, 4 USED ON P20
1400−017	46−307819P4	1	On−Line Power	TRANSFORMER, 25VA, 415/380VIN, 40 OUT C.T., 4 USED ON P21
1625−002−01	46−307819P53	1	On−Line Power	BOARD, SCR DRIVER, 3 USED ON P20 AND P21
1625−003−04	46−307819P12	1	On−Line Power	BOARD, CONTROL LOGIC, 3 USED ON P20 AND P21
1625−004−001	46−307819P13	1	On−Line Power	BOARD, SCR SNUBBER, 18 USED ON P20 AND P21
1625−031−04	46−307819P15	1	On−Line Power	BOARD, SCR FAILURE SENSE, USED ON P20 AND P21
1625−059−01	46−307819P46	1	On−Line Power	BOARD, PDU CONTROLLER
1680−008	46−307819P36	1	On−Line Power	SWITCH, PUSHBUTTON, 3 USED ON P20 AND P21
1680−021	46−307819P19	1	On−Line Power	SWITCH, BYPASS 3 PH 18.5 KW 600V 55A, USED ON P20 AND P21
2075−002	46−307819P54	1	On−Line Power	FUSE, 1A 600V, 4 USED (NOT SHOWN)
2075−020	46−307819P28	1	On−Line Power	FUSE AND CAP, .5A 300V SLO BLO, 3 USED, USED ON P20 AND P21
2075−023	46−307819P43 ‘	1	On−Line Power	FUSE, .25A 500V DUAL ELEMENT SLO BLO, 5 USED USED ON P20 AND P21
2085−007	46−307819P48	1	On−Line Power	HOLDER, FUSE, FLAT MOUNT, 5 USED ON P20 AND P21
2085−014	46−307819P54	1	On−Line Power	HOLDER, FUSE, PANEL MOUNT, .5A 300V, 3 USED ON P20 AND P21
4000−108	46−307819P31	1	On−Line Power	WIRE, BUS (FUSELINK) 18 GAGE, USED ON P20 AND P21
730220022−01	46−307819P50	1	On−Line Power	FUSE, 2A 600V SLO BLO, 2 USED
731000002−01	46−307819P49	1	On−Line Power	HOLDER, FUSE, FLAT MOUNT, 2 USED
9100−279−01	46−307819P89	1	On−Line Power	CIRCUIT BREAKER KIT
9000−268−01		1	On−Line Power	SHUNT−TRIP PROTECTION ASSY

3−5−1 VENDOR PARTS LIST USED IN LIGHTNING PDU (continued)

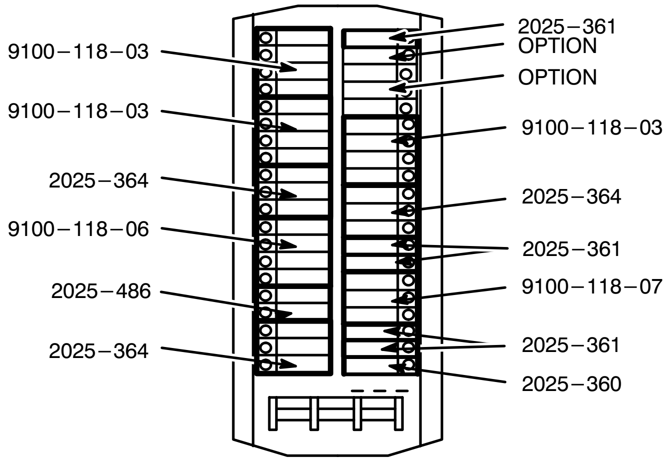


LIGHTNING PDU
ILLUSTRATION 3−19

3–5–1 VENDOR PARTS LIST USED IN LIGHTNING PDU (continued)

Vendor No.	GE Part NO.	FRU	Vendor	Description (Remarks)
1200–059	46–307819P69	2	On–Line Power	TERMINAL BLOCK, 600V, AWG# 3, UK16N, 16 USED
1200–061	46–307819P70	2	On–Line Power	TERMINAL BLOCK, 600V, AWG# 8, UK 10, 13 USED
1200–079	46–307819P71	2	On–Line Power	TERMINAL BLOCK, #2 AWG, 9 USED
1200–084	46–307819P72	2	On–Line Power	TERMINAL BLOCK, #4 AWG TO 3/0
1360–005	46–307819P77	2	On–Line Power	STANDOFF, NYLON, HEX, 1 in. x 6–32, 24 USED, USED ON P20 AND P21
1400–018	46–307819P5	2	On–Line Power	TRANSFORMER, CONTROL, 1VA, 40VIN, 9.6VOUT, 50/60 HZ, 3 USED, USED ON P20
1400–019	46–307819P6	2	On–Line Power	TRANSFORMER, CONTROL, 1VA, 40VIN, 9.6VOUT, 50/60 HZ, 3 USED, USED ON P21
1450–954	46–307819P108	N	On–Line Power	3 PH 50 KVA 480 VIN 208Y/120 VOUT 60 Hz, USED ON P20
1450–955	46–307819P109	N	On–Line Power	3 PH 50 KVA 380/400/415 VIN 208Y/120 VOUT 50 HZ, USED ON P21
1600–020	46–307819P10	2	On–Line Power	LED(RED) 6V, 10 mA, USED ON P20 AND P21
1660–003	46–307819P17	1	On–Line Power	RELAY 24VDC 10A, USED ON P20 AND P21
1660–004	46–307819P79	2	On–Line Power	RELAY SOCKET, USED ON P20 AND P21
1660–022			On–Line Power	RELAY SOCKET, USED ON P20 AND P21
1660–053			On–Line Power	RELAY, 120VAC, 10A, USED ON P20 AND P21
1680–005	46–307819P60	2	On–Line Power	COVERS FOR EPO SWITCHES, 3 USED ON P20 AND P21,
1680–007	46–307819P62	2	On–Line Power	LENSES FOR EPO SWITCHES YELLOW, 2 USED ON P20 AND P21,
1680–074	46–307819P61	2	On–Line Power	LENS, EPO, WHITE
2085–015	46–307819P68	2	On–Line Power	FUSE CLIP, USED FOR #2075–020, 3 USED, USED ON P20 AND P21
3050–081	46–307819P65	2	On–Line Power	RECEPTACLE L21–30R1, 2 USED
3050–113	46–307819P58	2	On–Line Power	TEST JACK RED, 7 USED ON P20 AND P21
5100–023	46–307819P74	2	On–Line Power	BUS BAR
5100–114	46–307819P74	1	On–Line Power	NEUTRAL AND GROUND BUS BAR
5100–032	46–307819P75	2	On–Line Power	TRANSFORMER STRAP, 2 USED
7060–005	46–307819P33	2	On–Line Power	12 in. LOGIC TO DRIVER BOARD HARNESS, 3 USED, USED ON P20 AND P21 (NOT SHOWN)
7060–012	46–307819P34	2	On–Line Power	6 in. SCR HARNESS 22/4X6, 18 USED, USED ON P20 AND P21 (NOT SHOWN)
7060–089	46–307819P67	2	On–Line Power	MAIN HARNESS ASSEMBLY USED ON P20 AND P21 (NOT SHOWN)
9100–050	46–307819P78	2	On–Line Power	ASSEMBLY, THERMAL SWITCH, USED ON P20 AND P21

3–5–2 Circuit Breaker Panel



CIRCUIT BREAKER PANEL
ILLUSTRATION 3–20

TABLE 3–3
CIRCUIT BREAKER CONFIGURATION FOR MODELS P20 AND P21

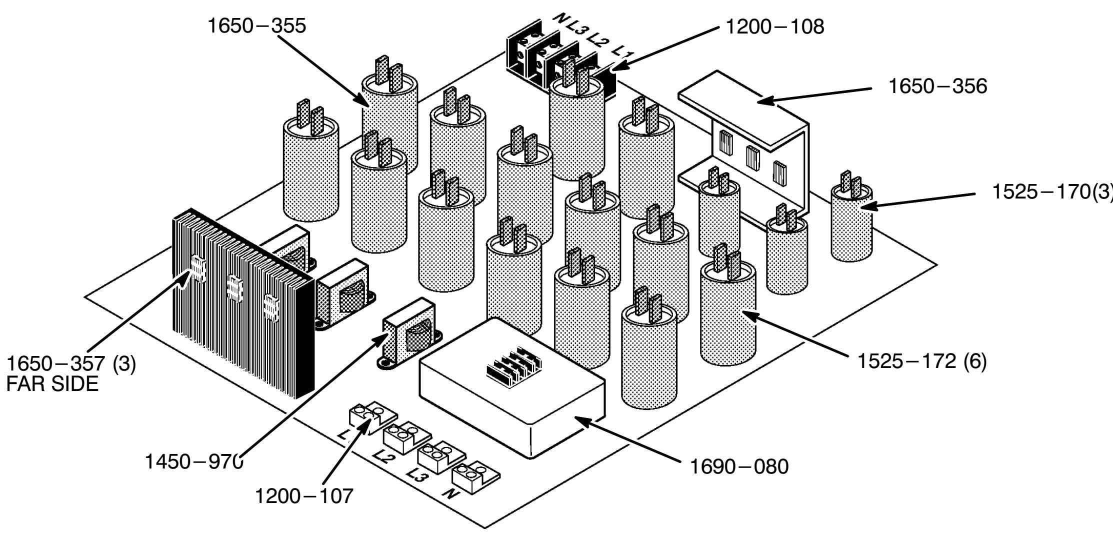
PART NO.	FUNCTION	BREAKER SIZE USED	FEEDER SIZE	OUTPUT TERMINAL STRIP NO (A2) TB10	POLES
9100–118–03	GRADIENT AMPLIFIER CABINET 1	60A	60A	1, 2, 3	3W/SHUNT TRIP
9100–118–03	GRADIENT AMPLIFIER CABINET 2	60A	60A	4, 5, 6	3W/SHUNT TRIP
2025–364	SUPERCON SHIM POWER SUPPLY	30A	35A	33, 34, 35,	3
9100–118–06	RF AMPLIFIER	30A	35A	10, 11, 12	3W/SHUNT TRIP
2025–486	CUSTOMER OPTIONS	15A	35A	13, 14	2
2025–364	SYSTEM CABINET	30A	35A	16, 17, 18	3
2025–361	HEAT EXCHANGER	15A	25A	20	1
NONE	SPARE	NONE	25A	21	1
NONE	SPARE	NONE	35A	22, 23, 24	3
9100–118–06	GRAM CABINET	30A	60A	7, 8, 9	3W/SHUNT TRIP
2025–364	MAGNET POWER SUPPLY	30A	35A	36, 37, 38	3
2025–361	OPERATOR WORKSPACE (PC)	15A	45A	31	1
2026–361	SERVICE OUTLET	15A	25A	32	1
9100–118–07	GRADIENT WATER CHILLER*	15A	25A	26, 27	2W/SHUNT TRIP
2025–361	MFC	15A	25A	19	1
2025–361	PDU POWER	15A			1
2025–360	OPERATOR WORKSPACE (HOST)	20A	25A	25	1

* NOTE: On earlier versions of the P20 and P21, this function was optional Resistive Shim P.S. which also used Part No. 9100–118–07.

3–5–2 Circuit Breaker Panel (continued)

Vendor No.	GE Part NO.	FRU	Vendor	Description (Remarks)
2025–361	46–307819P42	1	On–Line Power	BREAKER, CIRCUIT 1P 15A 120VAC, 5USED
2025–364	46–307819P40	1	On–Line Power	BREAKER, CIRCUIT 3P 30A 240VAC, 3 USED
2025–486	46–307819P	1	On–Line Power	BREAKER, CIRCUIT 2P 15A 240VAC, 1 USED
9100–118–03	46–307819P82	1	On–Line Power	BREAKER, CIRCUIT 3P 60A 240VAC 24VDC SHUNT
9100–118–07	46–320337P4	1	On–Line Power	BREAKER, CIRCUIT 2P 15A 240VAC 24VDC SHUNT
9100–118–06	46–320337P1	1	On–Line Power	BREAKER, CIRCUIT 3P 30A 240VAC 24VDC SHUNT

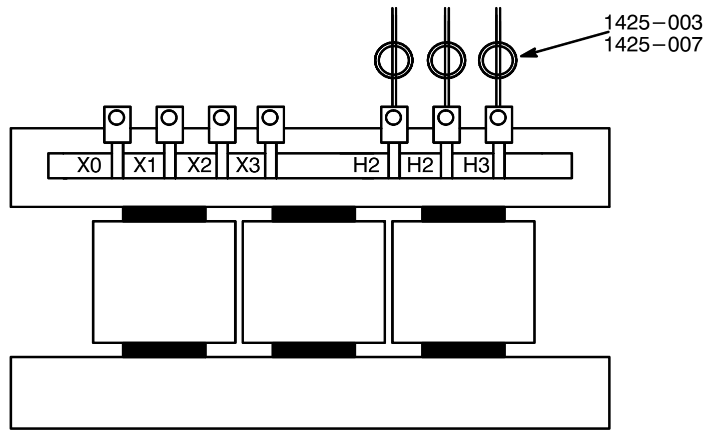
3–5–3 Filter Assembly Panel



FILTER ASSEMBLY
ILLUSTRATION 3–21

Vendor No.	GE Part NO.	FRU	Vendor	Description (Remarks)
1690–080	46–307819P95	1	On–Line Power	ACCUVAR, ACV–Y
1200–107	46–307819P96	5	On–Line Power	LUG, 2 POSITION, #14–2/0 RANGE
1200–108	46–307819P97	1	On–Line Power	TERMINAL STRIP, 30 AMP
1525–170	46–307819P98	3	On–Line Power	CAPACITOR, UL, 10.0µf, 660 V
1530–009	46–307819P99	3	On–Line Power	CAPACITOR BRACKET, FOOTED
1525–171	46–307819P100	6	On–Line Power	CAPACITOR, UL 80.0µf, 240 VAC
1530–010	46–307819P101	12	On–Line Power	BRACKET 2.5 RND, STAND–UP
1525–172	46–307819P102	6	On–Line Power	CAPACITOR, VDE, 45µf, 370 VAC
1450–970	46–307819P103	3	On–Line Power	INDUCTOR, SERIES 15A, 500 uh
1690–079	46–307819P104	3	On–Line Power	MOV 320 V, 20MM, 6.5 KA (Z321162UL)
1650–355	46–307819P105	15	On–Line Power	RESISTOR, 220 Kohm, 2W, MO
1650–356	46–317819P106	3	On–Line Power	RESISTOR, 2 ohm, 25W, WIRESOUND
1650–357	46–317099P107	3	On–Line Power	RESISTOR, 1.0 ohm, 50W, WW

3−5−4 Current Transformer, Models 46−317099 P20 AND P21



CURRENT TRANSFORMER
ILLUSTRATION 3−22

Vendor No.	GE Part NO.	FRU	Vendor	Description (Remarks)
1425−003	46−307819P8	1	On−Line Power	TRANSFORMER, CURRENT 100:5, 3 USED, USED ON P20
1425−007	46−307819P9	1	On−Line Power	TRANSFORMER, CURRENT 200:5, 3 USED, USED ON P21

3−6 PERIODIC MAINTENANCE

3−6−1 When to Perform

Perform Periodic Maintenance (PM)

- at time of installation.
- each sixth month after installation in a mobile site.
- each twelfth month after installation in a fixed site.

3−6−2 How to Perform

All periodic maintenance can be accomplished through outer door and load panel. However, time required for completion may be reduced if at least one side or rear panel is also removed.

Perform a complete cleaning and inspection as described in Section 3−6−4, PM Procedure. Three hours are required to perform this procedure.

3−6−3 Tools Required

Common electrician tools are sufficient to complete this procedure, including torque wrench with allen sockets.

A recording torque wrench is needed if left side panel cannot be removed.

A vacuum cleaner with hose may be required.

3−6−4 PM Procedure

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN LIGHTNING PDU EVEN WHEN ALL LIGHTNING PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Lightning PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points − must measure 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.

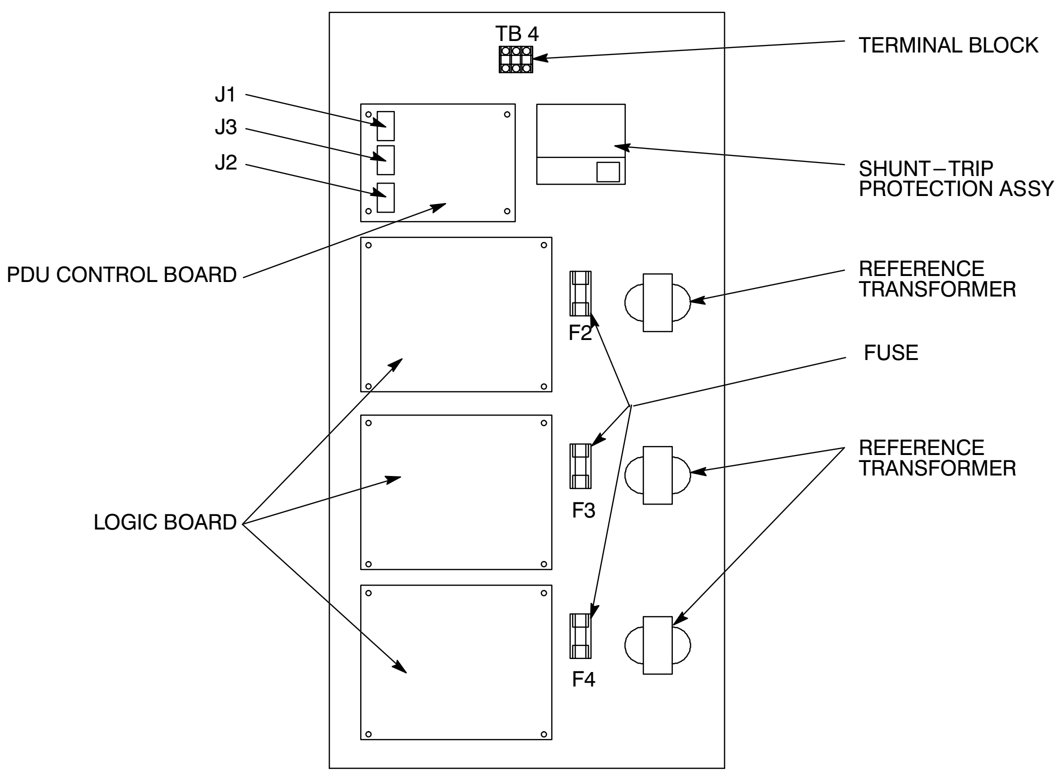
3–6–4 PM Procedure (continued)

6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove and keep two nuts securing articulated arm to top of dead front.
8. Close dead front.
9. Remove and keep 10 screws securing dead front to load panel. Remove dead front covering load panel.
10. Check all circuit breaker connections for tightness. Torque to 30 lb in (34 Nm).
11. Inspect for dust. Vacuum any dust accumulation.
12. Put dead front on load panel and secure with 10 screws kept in Step 9.

Note

Verify that dead front will fit on the two guides at right when load panel door is closed.

13. Open load panel door. Install two nuts securing articulated arm to top of dead front.
14. Lock articulated arm.
15. Inspect circuit boards located on inner surface of load panel circuit breaker compartment and on regulation panel. See Illustrations 3–23 and 3–24.



LOAD CENTER BRACKET ASSEMBLY (A14)
ILLUSTRATION 3–23

3−6−4 PM Procedure (continued)

a. Inspect for evidence of overheating. Inspect for cracked, burned or distorted resistors, capacitors, and charred insulation. Replace any overheated parts.

b. Inspect for evidence of physical damage such as broken or cracked capacitors and resistors, broken solder joints and discontinuity in circuit foil paths.

c. Inspect circuit board connectors for corrosion.

d. Replace board if damaged. Refer to Section 3−4−1, Service a Circuit Board.

16. Inspect and torque all input power connections on SCRs, circuit breakers, switches, transformers, and filters. Inspect for discoloration or any other sign of excessive heating. Torque power connections as follows:

    a. Torque input power connections (eight screws) to 30 lb in (34 Nm).

    b. Torque output power connections to 30 lb in (34 Nm).

    c. Torque six main circuit breaker connections to 25 lb in (113 Nm). See Illustration 3−25.

    d. Torque nine power wires on bypass switch. See Illustration 3−25. Torque to 30 lb in (34 Nm)

Note

With left side panel in place, use a recording torque wrench.

    e. Torque X_0, X_1, X_2, X_3 and H_1, H_2, H_3 to 30 lb in (34 Nm).

    f. Torque filter terminal block to 30 lb/in (34 NM) connections (A11A3, A11A4).

    g. Torque phillips screws on all 18 SCRs (54 screws) to 30 lb in (34 Nm). See Illustration 3−24.

17. Check connectors on circuit boards to insure proper seating. See Illustrations 3−23 and 3−24. This includes:

    a. Connectors on both ends of 18 snubber board harnesses.

    b. Connectors on both ends of 6 logic board/driver board harnesses.

    c. Nine connectors on SCR sense board.

    d. Three connectors on PDU control board.

18. Check that relay A9K1 is properly seated. See Illustration 3−24 for location of relay.

19. Periodic maintenance for fan assembly (A7):

    a. Remove and keep four screws securing fan panel assembly to frame. Remove fan panel assembly.

    b. Remove and keep fan mounting screws. Remove fan guard and fan.

    c. Remove aluminum fan filters. Clean and install.

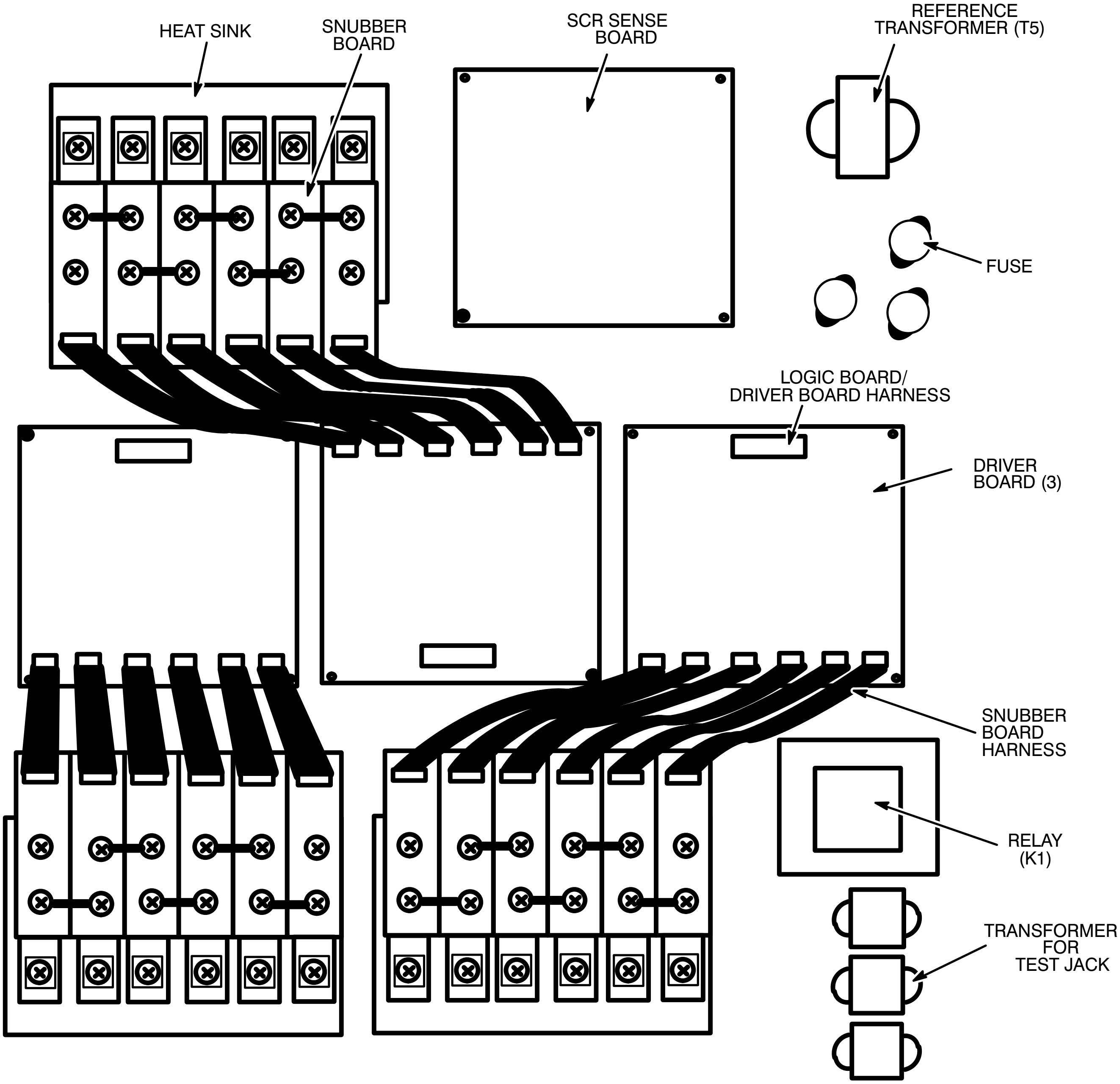
3−6−4 PM Procedure (continued)

Note
Mount fan so air flow is into Lightning PDU.

d. Check that fan rotors are clean and rotate freely.

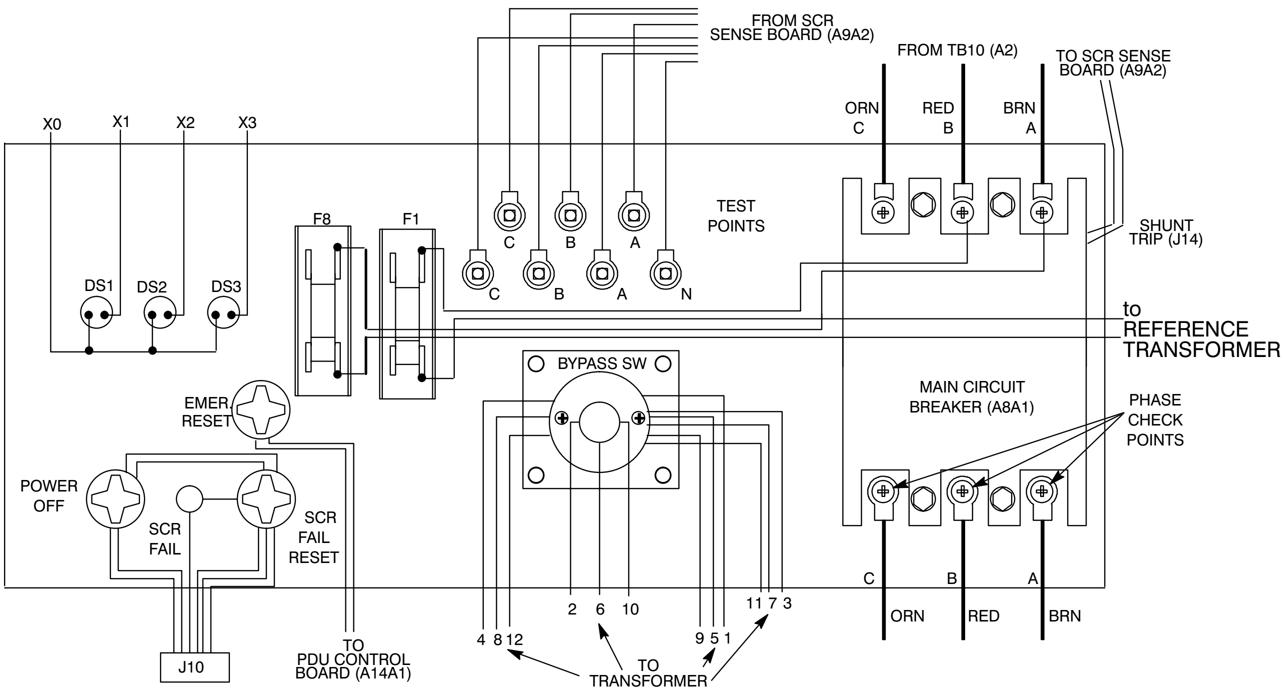
e. Install fan guard and fan. Secure with fan mounting screws kept in Step 19−b.

f. Install fan panel assembly and secure with four screws kept in Step 19−a.



REGULATION PANEL
ILLUSTRATION 3−24

3−6−4 PM Procedure (continued)



FRONT PANEL (REAR VIEW)
ILLUSTRATION 3−25